

Honeywell

Communication Link Module (CLM)

**CLM/Applied Automation
Optichrom Advance System
(HCI-A) Interface
User Manual**

SST-5D0

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About This Document

This document is the manual for the Communication Link Module (CLM) interface to the Applied Automation Optichrom Advance System using the HCI-A protocol.

The CLM/HCI-A interface can be hosted on two different platforms. Both are completely supported by this interface manual.

- CLM/HCI-A is provided for the original CLM which is hosted on an Application Module (AM) and requires the CLMI or CCP communications boards.
- APP/CLM HCI-A installs to the Honeywell Application Processing Platform (APP) using the APP Base Software (AM personality) and actually emulates the CLMI/CCP communications boards used in the AM version of the CLM. The LCN software and user station operating displays have not changed from the AM-based CLM and are installed in the same manner. The board emulation and protocol software is installed in the Windows environment of the APP, and therefore requires additional software and installation and configuration procedures. The basic functionality and operation of the HCI-A interface is the same.

Both versions of the HCI-A application package are provided together on two media, one for the TPS (Windows 2003-based) or Experion (Windows 2008 based) files that are required for the APP/CLM and one for the LCN-based files. Installation procedures determine the version to be implemented. Note that if you order the CLM version, you will get the TPN (LCN) media only.

The CLM provides a standard communications platform, which allows the TPS or the Experion to interface with a wide variety of devices and networks. With this interface, the TPS or the Experion Station operator has access to points located on a non-TPS network.

The complete CLM functionality is segmented into the following four logical layers:

- Process Application
- CLM Platform
- Network Interface
- Communication Protocol

The bottom three logical layers provide the communications foundation for the top layer or Process Application layer. The Process Application layer is the actual user application and is not considered part of the CLM Platform.

About This Document

This document is primarily concerned with the bottom two logical layers: Network Interface and Communication Protocol, with a specific orientation to interfacing the Applied Automation Optichrom Advance System to the TPN using the HCI-A protocol. This document supplements the *CLM Platform Implementation Manual*, which covers the CLM Platform logical layer, and the portion of the Network Interface logical layer that is not oriented to a specific non-Honeywell network.

This document is primarily intended for the engineer who will configure the interface between the TPS or Experion and an Applied Automation Optichrom Advance System. The document also includes information that supplements the *CLM Platform Implementation Manual* with specific details regarding the Applied Automation Optichrom Advance System network.



ATTENTION:

With CLM R200.1, CLM is supported on Experion APP (Windows 2008 platform on both 32 bit and 64 bit variants). Please note that CLM R200.1 is only meant for Windows 2008 nodes (Experion PKS R400/R410). CLM R200.1 does not work on Windows 2003.

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.

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1. Interface Overview

1.1 Manual Description

Introduction

This manual describes how to interface a Communication Link Module (CLM) to an Applied Automation Optichrom Advance System device using the HCI-A protocol. The CLM may be hosted on an AM node (AM/CLM) or an APP (APP/CLM).



ATTENTION:

With CLM R200.1, CLM is supported on Experion APP (Windows 2008 platform on both 32 bit and 64 bit variants). Please note that CLM R200.1 is only meant for Windows 2008 nodes (Experion PKS R400/R410). CLM R200.1 does not work on Windows 2003.

What this manual does

This manual documents the CLM network interface layer that is used to configure a CLM as an Applied Automation Optichrom Advance System device. The manual includes the LCN image points used to reflect the data from the Applied Automation Optichrom Advance System analyzers. This document also includes information that supplements the *CLM Platform Implementation Manual* with specific details regarding the Applied Automation Optichrom Advance System interface using HCI-A protocol.

What it does not do

This manual does not document how to write specific process applications using the Applied Automation Optichrom Advance System analyzer data. Refer to the *CLM Platform Implementation Manual* for any ancillary information that is required to complete a CLM/ Applied Automation Optichrom Advance System interface implementation. Refer to the appropriate vendor manuals for process application information.

Intended audience

This document is primarily intended for the Network Interface Design Engineer, who has the responsibility for definition of an Applied Automation Optichrom Advance System interface, and its process data for use by the TPS or the Experion Operator and the Process Engineer.

Prerequisite knowledge

Since this manual supplements the *CLM Platform Implementation Manual*, you should become familiar with the CLM Platform manual first, especially the following sections:

1. Interface Overview

1.2. Overview

- Defining Interface Structure Templates
- Building CLM Interface Structures
- Network Interface Design Considerations

1.2 Overview

Introduction

This section provides an overview of the CLM/Applied Automation Optichrom Advance System interface. CLM refers to the CLM and protocol software. The CLM may be hosted on an AM node (AM/CLM) or an APP (APP/CLM). In this manual, CLM is used to describe both the AM/CLM and the APP/CLM.

AM/CLM is a modified AM Node

The AM/CLM is hosted on an AM node. The AM is modified to contain a CLMI board and a CLM paddle board set.

APP/CLM is an APP

The APP/CLM is hosted on an APP and contains no special CLMI or paddle card set. The APP/CLM emulates the CLMI with software, and uses the PC serial communication ports instead of the CLM paddle card. An optional port expander may be purchased to provide additional communication ports.

AM/APP requirements

The AM must be R600 or later revision with at least 4-megawords of memory.

The APP must be R400 and contain an LCNP board.

CLM Communication boards

The CLM communication board may be either a Communication Link Module Interface (CLMI) board or a Communication Control Processor (CCP) board. The CLM may contain a maximum of two sets of CLM/CCP communication boards and their respective paddle boards.

The APP/CLM does not contain a physical CLMI/CCP communication board. The CLMI/CCP board is simulated via software in the APP. The APP/CLM supports two emulated boards.

Port communications

The Applied Automation Optichrom Advance System analyzers that form the Advanced Data Hiway connect to an HCI-A card that acts as a bridge to the CLM. The Applied Automation Optichrom Advance System analyzer network's HCI-A board is physically connected to the CLM via an RS 232 link to one port.

The CLM HCI-A application supports four ports. In an AM/CLM, the paddle board has four communication ports. In a APP/CLM, the PC serial ports are utilized. The APP/CLM normally supports up to two serial connections. An additional PC card may be purchased to allow the APP/CLM to support up to eight serial connections.

The CLM HCI-A protocol supports point-to-point connections from an HCI-A board, and multidrop connections from Advance Data Hiways (Refer to figure 1.2-1).

Be sure to check the APP/CLM Planning and Installation Manual for cabling specifications for the AM/CLM and APP/CLM.

Point capacity

The approximate point capacity of a 4-megaword AM/CLM is 2000* image points. A 6-megaword AM/CLM will double this capacity to 4000* image points.*

AM/APP point defines image point

The CLM interface provides the TPS and the Experion operator access to the Applied Automation Optichrom Advance System analyzers' streams. The CLM uses AM custom points, with Custom Data Segments (CDSs), to create image points that mirror information in the analyzer.

Parameter changes by TPS operator

The Universal Station operator can monitor and change selected operating parameters in the Applied Automation Optichrom Advance System analyzer by changing the associated parameters in the CLM image point.

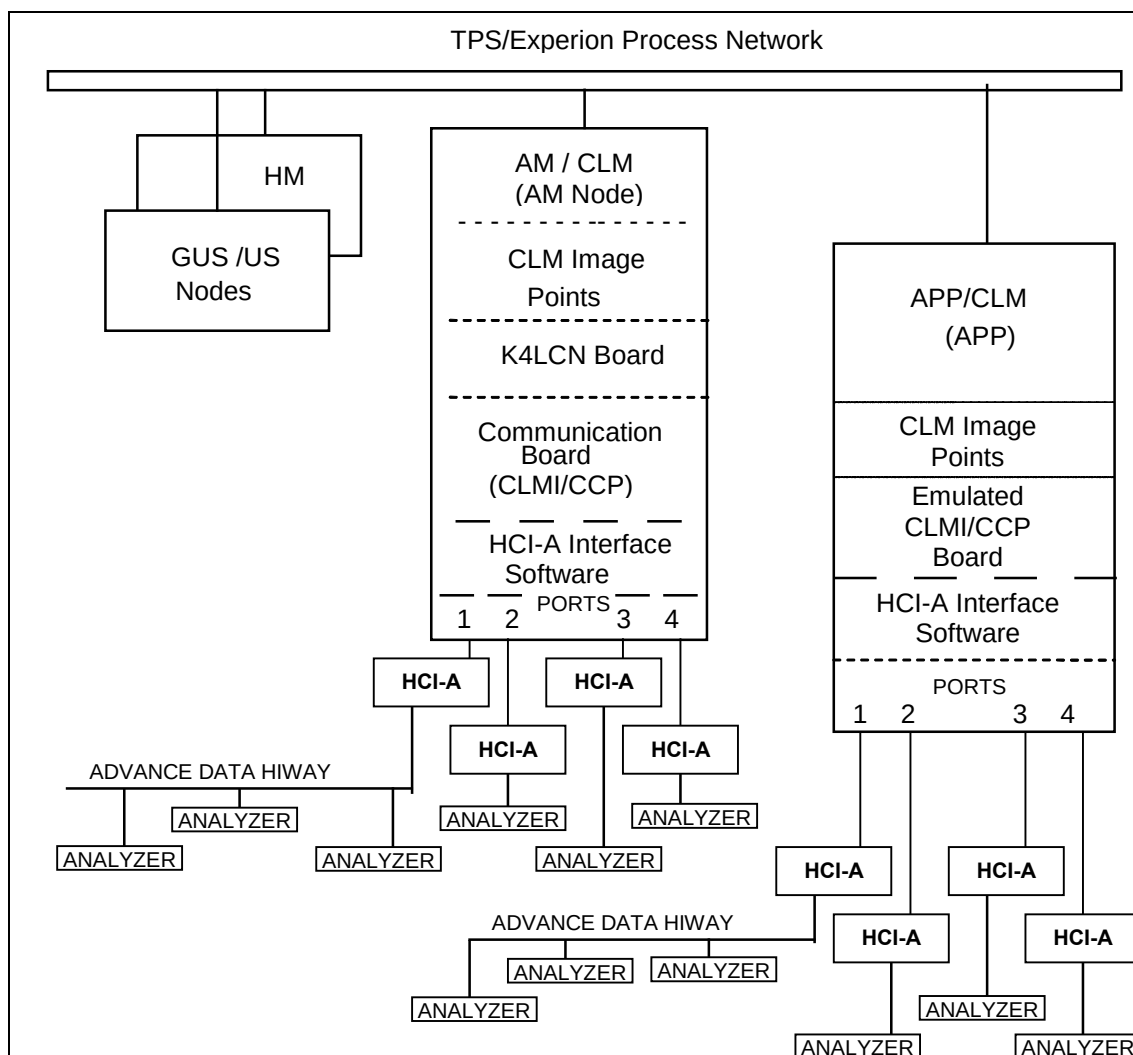
CLM Interface Overview

Figure 1.1 presents an overview of the interface between the Applied Automation Optichrom Advance System analyzer and the TPS or Experion Process Network.

* It should be recognized that it is unlikely this interface will be configured with these maximum point counts.

1.2. Overview

Figure 1-1 Overview of the CLM Interface



Supported Features

The interface between the Applied Automation Optichrom Advance System analyzer and the TPS or Experion Process Network supports the following features.

- Allow Operator to set the clocks of all network analyzers.
- Allow Operator to activate an Event at a selected analyzer.
- Indications of Analyzer, Stream, and Component status.
- Accept analyzer results on an “as available” basis and store results to AM image points.

1.3 Communications Characteristics

Communications Characteristics

The following table depicts the protocol specifications of the HCI-A protocol.

Table 1-1 HCI-A Specification

Characteristic	Requirement
Protocol	HCI-A, ASCII
Coding System	1 Start bit, 7 Data bits, 1 Parity bit (Even), 2 Stop bits,
Connection Type	Direct connection between CLMI and HCI-A board. (Multi-drop from Advance Data Hiway)
Communication Media	Wire, RS-232C Half Duplex
Configurable Baud Rates	300 to 9600 bps
Configurable Timeout	Default at 2 seconds, incremental by 1 second up to 5.
Configurable Retry's	Default at 3
Data Acquisition	Accept analyzer results on an unsolicited basis.

1.4 Protocol Messages

HCI-A Protocol messages

The following HCI-A protocol messages are supported for this interface. Note that the numeric value is hexadecimal.

- - OK, Message Acknowledgment
- - NO, Request Not Accepted
- - TOR, Transmission of Results
- - HAE, Activation of an Event and Request for Results
- - TAD, Time and Date Set
- - PARM, Selection of HCI-A Interface Parameters
- - RIP, Request for Interface Parameters

1.5 Physical Connection

Physical connection

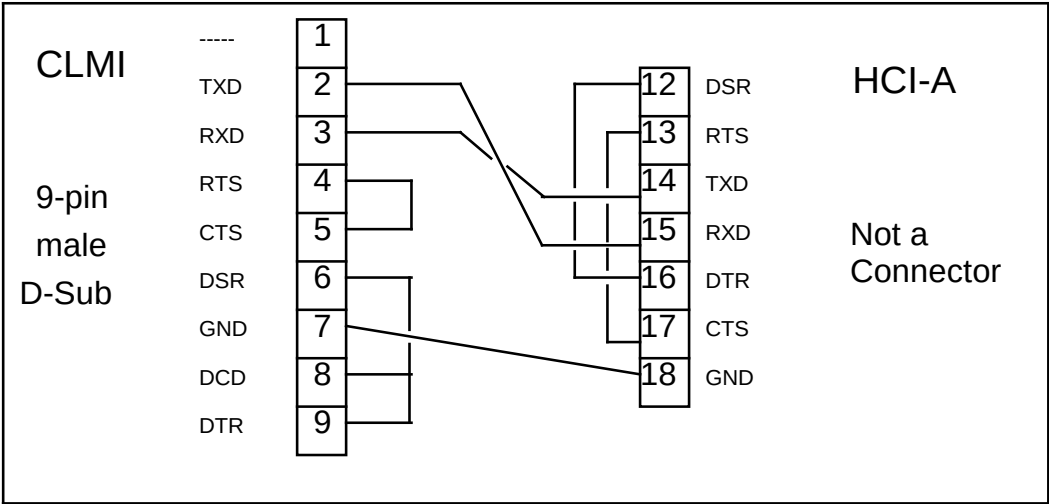
The physical connection is an RS-232 link between the HCI-A card in the Applied Automation Optichrom Advance and the CLM. It may be as simple as a short direct connect or null modem cable to a single local analyzer, a connection through modems, or a multidrop connection utilizing an Advance Data Hiway.

The CLM connection will be to a port on the paddle card for an AM/CLM, or to a PC serial port for a APP/CLM.

AM/CLM Cable Description

The port connection at the CLMI in the AM/CLM requires a 9-pin male connector. The HCI-A is connected using the RS-232C serial line wiring without handshaking. Note, HCIA pin 12 is tied to HCIA pin 16 and HCIA pin 13 is tied to HCIA pin 17. Figure 1.2 depicts the RS-232 connection between the CLMI daughter card and the HCI-A.

Figure 1-2 CLMI to HCI-A Cable Description



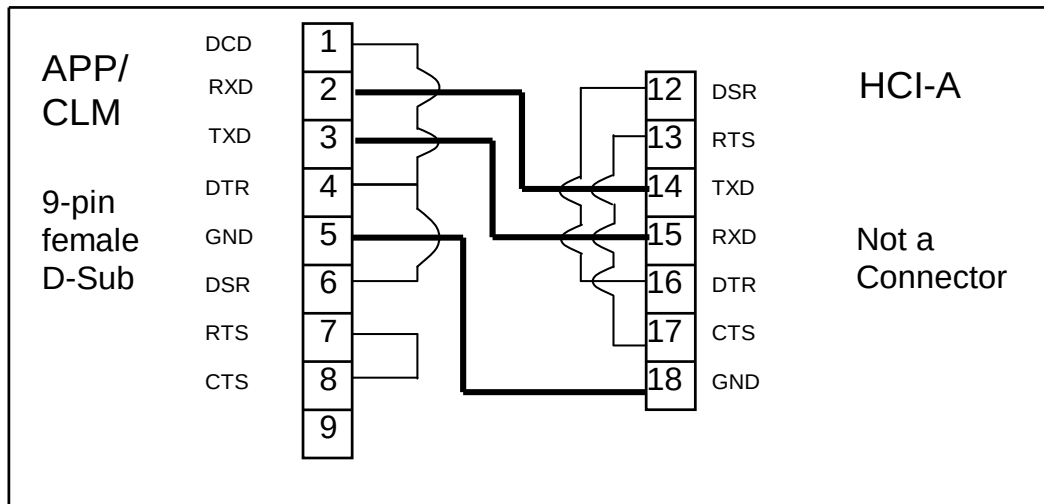
APP/CLM Cable Description

The standard PC serial port connection at the APP/CLM requires a 9-pin male connector. The HCI-A is connected using the RS-232C serial line wiring without handshaking. Figure 1.3 depicts the RS-232 connection between the APP node (via standard PC serial port) and the HCI-A.

1. Interface Overview

1.5. Physical Connection

Figure 1-3 APP/CLM to HCI-A Cable Description



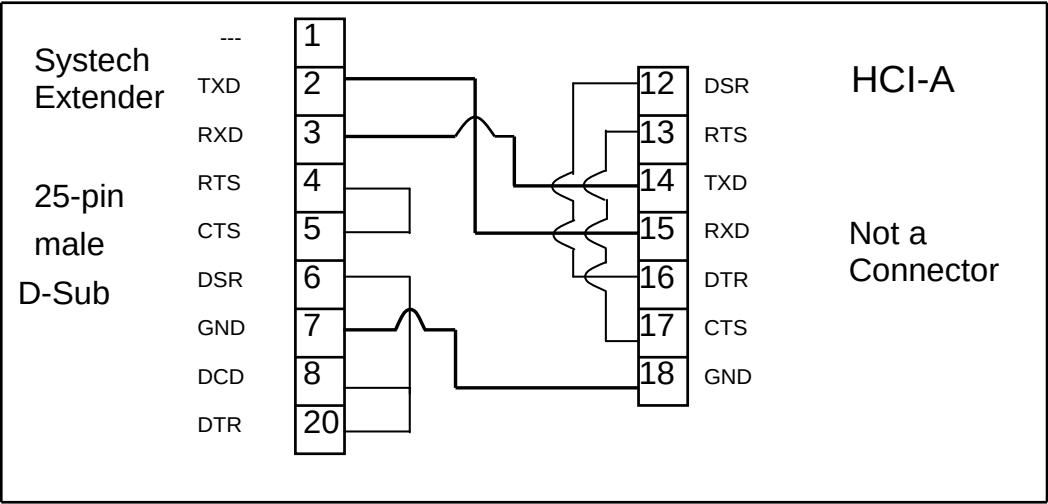
Optional APP/CLM Port Expander Cable Description (for Systech Extender)

The APP/CLM has an optional port expander (COM port systech extender) to allow implementation of the eight maximum configurable ports. The COM port systech extender connection requires a 25-pin male connector. Figure 1.4 depicts the RS-232 connection between the optional COM port systech extender and the HCI-A.

Note: CLM is supported on Windows 2003. The SYSTECH COM Port Extender needs to be used in place of Stallion card.

With CLM R200.1, CLM is supported on Experion APP (Windows 2008 platform on both 32 bit and 64 bit variants). Note that CLM R200.1 is only meant for Windows 2008 nodes (Experion PKS R400/R410).

Figure 1-4 Systech Extender to HCI-A Cable Description



1. Interface Overview
1.5. Physical Connection

2. Communication Link Module

2.1 Communication Link Module Logical Structure

Introduction

The CLM interface supports a user application that may monitor or provide control information to the process.

Four CLM layers

The Communication Link Module consists of four logical layers. The following is a brief definition of those layers.

Process application layer

The process application layer represents the actual application that is exchanging data with the analyzer device. It consists of process application points, Control Language programs, and user displays that are built to monitor and control a process. The process application is built on top of the other three layers, and is the end object of the CLM interface.

CLM platform layer

The CLM platform layer is the common layer that exists in all CLM environments, and consists of both software and hardware. The platform provides a base for building a gateway to the TPS or Experion Process Network. The physical connection is an RS-232 link between a CLMI board in the AM/CLM (or PC serial port in the APP/CLM) and the analyzer (device) network.

Network Interface layer

The Network Interface layer defines the type of network and devices connected. It includes the image points used to reflect the data. No conversion codes are used with the HCI-A interface because this protocol communicates all data in ASCII format. This Network Interface layer is the subject of this user's manual. Refer to the *CLM Platform Implementation Manual* for additional information required to complete the implementation.

Communication protocol layer

The communication protocol moves the data between the network interface device and the CLM image point for display and use by the upper layers.

2. Communication Link Module

2.1. Communication Link Module Logical Structure

3. Data Structures

3.1 Introduction

Introduction

This interface requires CLM data structures to define network structure and point data values.

3.2 Building the Interface

Introduction

All CLM interface structures are defined using the CLM Configuration Builder, which is part of the CLM Platform software. These interface structures include the following:

- Network Definition Table (NDT)
- Point Interface Table (PIT)
- Point Interface Table Entry (PITE)

A complete description of the CLM Configuration Builder and the build process is included in the CLM Platform Users Manual. However, this section includes a summary of the steps required to complete the definitions of the interface structures.

Builder

The CLM Configuration Builder (referred to as the Builder) accepts input definition from a file or on-line from the Universal Station keyboard. The input file mode requires a Builder Input File (BIF) for each interface structure being defined.

BIF files on the HM

By default, the builder looks for BIF files in the NET>&CUS directory. However, the user may place BIF files in a user-named directory. If a different directory is used, the directory name is required in front of the file name when you enter it in the builder (for example, USER>BIF.ZI).

BIF files created before actual build process

BIF files may be created in advance using the LCN file editor or any other text file editor. The BIF files are then transferred to the HM directory stated above for use by the Builder during the actual build process. Refer to **Sections 6 and 7** of this document for additional information on the BIF files.

3. Data Structures

3.2. Building the Interface

BIF validation with TDF

During the actual interface structure build process, the Builder validates each BIF input file using a pre-defined Template Definition File (TDF) for the specific interface structure type. These TDFs must be available in the HM directory &CUS for exclusive use by the Builder.

Building Network Data Structures

Define NDT structure

The following steps describe the creation of the BIF files required for your CLM interface. This will define and generate the Network Definition Table (NDT) network interface structures

Table 3-1 Define and Generate NDT Network Interface Structures

Step	Action								
1	Identify each addressable component in the network. The number of addressable network components will vary from site to site; however, all networks will have the following types.								
<table><tr><th><i>Component</i></th><th><i>Description</i></th></tr><tr><td>CLMI</td><td>CLMI Board, 1 or 2 per AM/CLM. Only 1 emulated CLMI per APP/CLM.</td></tr><tr><td>PORT</td><td>1 for each HCI-A card</td></tr><tr><td>DEVICE</td><td>1 for each network device</td></tr></table>		<i>Component</i>	<i>Description</i>	CLMI	CLMI Board, 1 or 2 per AM/CLM. Only 1 emulated CLMI per APP/CLM.	PORT	1 for each HCI-A card	DEVICE	1 for each network device
<i>Component</i>	<i>Description</i>								
CLMI	CLMI Board, 1 or 2 per AM/CLM. Only 1 emulated CLMI per APP/CLM.								
PORT	1 for each HCI-A card								
DEVICE	1 for each network device								

Step	Action
2	Create a BIF to include one NDT entry for each of the identified addressable network components. The BIF may be created using the LCN file editor, or other text file editor, and transferred to the History Module (HM) for use. Names of the BIFs included by Honeywell are listed in Table 3.4. Sample BIF listings for the CLMI, Port and Device are included in Section 6.2. Copy the appropriate BIF file, and complete each definition within the BIF, to create the BIF NDT entry for each addressable network component.
3	Build the NDT entries on the target CLM node. This is performed using the CLM Configuration Builder, which is installed on the CLM node as part of the CLM platform software. Instructions for using the CLM Configuration Builder are detailed in the <i>CLM Platform Implementation Manual</i>.

Building Point Interface Structures

Introduction

The Point Interface Table (PIT) interface structures are common across all CLM implementations. Therefore, instructions for defining BIF files for these interface structures, and prototype BIF listings, are included in the Platform manual; refer to the CLM Platform Implementation Manual.

Define PITE structure

Steps to define and generate the Point Interface Table Entry (PITE) interface structures.

3. Data Structures

3.3. Point Types

Table 3-2 Create Point Interface Table Entry (PITE)

Step	Action
1	Identify the specific data streams in the HCI-A network that require an AM image point.
2	Create a BIF for each of the identified data streams. * The BIF may be created using the LCN file editor, or other text file editor, and transferred to the HM directory of your choice (default is NET>&CUS) for use. Names of the prototype BIFs included by Honeywell are listed in Table 3.4. Sample BIF listings for each point type are included in Section 7.2. Copy the appropriate BIF file, and complete each definition within the BIF, to create the BIF PITE entry. Each PITE entry must be input to the CLM Configuration Builder to generate the PITE interface structure required for each data stream in your Applied Automation HCI-A network.
3	Build the image points and their point interface structures on the target CLM node. This is performed using the LCN Data Entity Builder and the CLM Configuration Builder, which is installed on the CLM node as part of the CLM platform software. Instructions for using the LCN Data Entity Builder are detailed in the LCN Data Entity Builder Manual. Instructions for using the CLM Configuration Builder are detailed in the <i>CLM Platform Implementation Manual</i>.

3.3 Point Types

Introduction

This interface requires both AM points and CLM points.

AM points

The following AM points are required.

Table 3-3 Required AM Point Types

Step	Required AM Point	Comment / Description
1	Dummy Regulatory Point	One point per CLM
2	CLM Analyzer Stream Status Point	One point per interface to support HCI-A graphics
3	Analyzer Stream Point	- Requires CL for full functionality - One point per stream
4	Stream Component Point	One point per stream component

AM: Dummy Regulatory point

One Dummy Regulatory point is required. This point is used by the Analyzer Stream point to define a local CDS array as having regulatory points as members of the array. Refer to **Section 7.2.1**.

AM: CLM Analyzer Stream Status point

One CLM Analyzer Stream Status point is required for the entire network to support the CLM Analyzer Stream Status display. Refer to **Section 7.2.2**.

AM: Analyzer Stream point

One Analyzer Stream point is required for each analyzer stream in the HCI-A network. Refer to **Section 7.2.3**.

AM: Stream Component point

One Stream Component point is required for each selected stream component. This point is an AM Regulatory point which must be designed and built by the Network Interface Design Engineer. Refer to **Section 7.2.4**.

3.4 Required Files for CLM / HCI-A Interface

This package has specific release level requirements for the TPS and Experion system and CLM Platform. Refer to the Customer Release Guide for release levels required. The Customer Release Guide also specifies History Module sizing requirements.

Note: This page does not include files included on the TPS media for the APP/CLM HCI-A version.

3. Data Structures

3.4. Required Files for CLM / HCI-A Interface

Table 3-4 Required Files For CLM / HCI-A Interface

Definitions / Requirements

HM NET

Directories	File Types
&CUS	.Z* , .S
UCLM	.EC, .CL, .AO, .DS

CLM Interface File Types and Extensions			
Extension	File Type	Extension	File Type
.AO	Control Language Object Modules	.ZC	CLMI Configuration Modification Sample
.CL	Control Language - Custom Data Segments (CDS)	.ZI	Builder Input File (BIF) Samples
.DS	Custom Display Schematics (Source)	.ZS	Status Display Files (CLMI/Port/device/etc.)
.EC	Executable Modules - Install,Link/Unlink CL	.ZT	Builder Template Definition Files (TDFs)
.S	CLMI Load Module		➤ Network Definition Table (NDT) Templates ➤ Point Interface Table Entry (PITE)

3. Data Structures

3.4. Required Files for CLM / HCI-A Interface

Package Files

CLM / HCI-A Files supplied on cartridge with interface:

* - File types EB, DS, and AO may be LCN Release dependent.
These files will be supplied for each supported release if required.

Install HCI-A to HM	HCI-A Protocol	Load Directive
EC	S	ZC
AA_INST	AAHCIA	AA_CLMI

Network Definition

ZT	ZI	ZS
AA_PORT	AA_PORT	AA_PORT
AA_DEV	AA_DEV	AA_DEV

CLM / HCI_A Operation Displays

DS*
CLMANLZA
CLMHCI_B

Point Image Files

ZT	EB	CL/CDS	CL/AO*
AA_STRM	AAPNTS	HCI_ASTRM	HCI_A_CL
ZI	OPTANLZ	ANLZMAIN	
AA_STRM	REGDUM		

Required CLM Platform Files

Required CLM Platform

ZC	ZT	ZI	ZS
CLMI	CLMI	CLMI	CLMI
	PIT	PIT	

Required CLM Platform Files by Point

CL/CDS
CLM_DMD3

File Definitions

Build Files	Desc.	AM Type	Requirements
AAPNTS	Stream Point	Switch	Image single stream from analyzer
	Component Pt.	Regulatory	Image single stream component
OPTANLZ	Stream Overview	Custom	Stream overview
REGDUM	Dummy Point	Regulatory	Array initialization

3. Data Structures

3.4. Required Files for CLM / HCI-A Interface

4. Install AM/CLM Network Interface Layer

4.1 Overview

Introduction (AM-Based CLM Only)

This section discusses how to install the CLM and HCI-A applications when using an **AM/CLM node**. The process for the installation of the network interface is covered within this section. At the appropriate point, it will require that you complete the installation of the CLM platform if you have not previously done so. After installation you will need to create the LCN image points to reflect your network and establish the connection between external network points and the LCN image points. (Refer to CLM Platform Implementation Manual, **Sections 6 and 7**.)

Use **Section 5** for instructions on how to install the HCI-A application when using a **APP/CLM node**.

4.2 Install HCI-A - Setup

Prior to installation

Review this package's Customer Release Guide.

WARNING

The automated installation procedure will query you to compile CDS packages. If the CDS's have already been installed and compiled then a recompilation will require that all points utilizing the CDS to be deleted and rebuilt. Recompilation of an existing CDS will change the date of the object file and this will cause a CDS mismatch to occur between points built with the original CDS and the newly compiled one.

Table 4-1 Pre-Installation Steps

Step	Action
1	Make a copy of the release volume.
2	Verify the path for the CL SOURCE/OBJ. We suggest that you use UCLM.
3	Verify the path for the CL CUSTOM GDF=CDSG. We recommend that only one network directory be used, and not changed
4	Set the CL PARAM LIST to the same path as the CL SOURCE/OBJ.

4. Install AM/CLM Network Interface Layer

4.3. Copy Install Files to HM

Step	Action
	This information is used when certain CDS modules are compiled
5	Set the USER DEFLT PATH to the same path as the CL SOURCE/OBJ This information is used when the supplied CDS modules are compiled and when the executable module link and unlink directives are executed.
7	Before the install procedure is executed, the paths must be identified and the volume of NET>&CUS> must exist. Normally, this directory is created during the installation of the AM. It was also required for installation of the CLM Platform product and should already be in place.□

Files included in the product release will be copied to your system during the install process.

4.3 Copy Install Files to HM

Copy files from distribution media

In the command function, within the engineering personality, execute the following EC and follow all of the prompts given within the EC procedure.

Step	Action	Result
1		The following Executes the AA_INST EC: Enter EC \$Fn>HCIA>AA_INST.EC \$Fn where n = the device identification for the drive (for example, \$F1)
2	A check is made to see if a certain file exists that would make this an upgrade. If this is a reinstallation, a message appears and gives user a chance to quit.	Answer Y and the EC will continue. Answer N and the EC will exit.

4. Install AM/CLM Network Interface Layer

4.4. Complete CLM Platform

Step	Action	Result
	DO YOU WISH TO CONTINUE?	
3	Are your paths acceptable?	Answer Y and the EC will continue. Answer N and the EC will exit.
4	The question what is the destination directory for the CL Source/Object.	Enter NET>UCLM where UCLM is the path name of the destination directory.
5	The question what is the destination directory for the custom display files.	Enter NET>UCLM where UCLM is the path name of the destination directory
6	The question what is the destination directory for the DATA BUILD files.	Enter NET>UCLM where UCLM is the path name of the destination directory
7		Copy other required files to NET>&CUS>. ZI, ZT, ZS, ZC, AAHCIA.S
8	If reinstallation, ask if want to re-compile the CDS. Do you want to continue compiles with overwrite switch?	If not reinstall then compiles CDS. If reinstall and N, then exit. If reinstall and Y, then compile CDS with overwrite switch.
	Note: If any errors exist, they must be corrected before continuing. Then the package will be resident on the destination LCN and installation procedures may continue. The Custom Displays included must be compiled in the desired directory. (Please see Schematics at 6.3)	

4. Install AM/CLM Network Interface Layer

4.4. Complete CLM Platform

Reinstallation / Upgrade

Refer to the Customer Release Guide for specific instructions on reinstallation. The install EC (AA_INST.EC) will prompt when specific action is required, such as adding the -OCD option when Custom Data Segments require reassembly.

4.4 Complete CLM Platform

Install Platform

Before proceeding, you must complete the CLM Platform installation if you have not already done so.

Note: This is required since the CLM and Builder points must exist in the CLM/AM database before you attempt to restart the node with the ZC file described in the next section.

4.5 Configure the CLMI / Port (CLM Only)

Communication protocol layer

The CLM Platform provides the lower level functionality to support the protocol layer on top of the platform. The protocol layer, developed by Honeywell, will include the software and other pieces required to communicate with the external devices. This protocol layer is the Applied Automation Optichrom Advance System HCI-A interface. This step defines the modification of the configuration information required by the platform to identify what hardware and software is used to accomplish communications with the external devices.

CLMI load directive file

Each AM node must have its own CLMI load directive file. This configuration file is an ASCII text file with the following naming convention.

- The name is the literal CLMI followed by three digit CLM physical node number and extension ZC (example: CLMI024.ZC)
- The file is stored in the NET>&CUS> directory
- For example, NET>&CUS>CLMI024.ZC is the name for physical node 24.

The CLMI load directive file creation is a part of the CLM Platform installation procedure. This implementation requires a modification of that directive file.

The form of the configuration directives to define the CLM environment are as follows.

Table 4-2 CLM (AM based): CLMI Load Directive

CLMI <board #>, <slot #> LOAD <file name> (dependent upon board type) INIT 0, <port map> RUN <entry offset> (dependent upon board type) END

CLMI slot numbers

Use the Maintenance Personality on a Universal Station to determine the CLMI slot numbers for each AM node.

LOAD File Name

The CLM communication board may be either a Communication Link Module Interface (CLMI) board or a Communication Control Processor (CCP) board. The name of the load file (.S) is dependent upon the type of board in the slot (CLMI.S or CCP.S).

Port Map

The port map is an eight-digit number. The first four digits correspond to the 4 CLMI ports. The second four digits must be zero. Enter a one to assign a port driver to a specific port. Enter a zero if the port is not used in this application.

Entry offset

The value of <entry offset> can change from release to release. Refer to the file AA_CLMI.ZC for the correct value.

Edit your configuration file

Next, add configuration information to the CLMI configuration file that was installed as part of the CLM Platform software.

4. Install AM/CLM Network Interface Layer

4.6. Adding another Network Interface

Step	Action
1	Set the current path name to NET>&CUS.
2	Using the text file editor, call up the text file NET>&CUS>CLMI0nn.ZC, where 0nn is the CLM physical node number. For example, CLMI024.ZC is the CLMI configuration file for CLM node 24.
3	The first 6 directives are always used. The following example configures the first board in slot five and port three. <pre>CLMI 1,5 LOAD <CLMI.S or CCP.S> RUN <41100 or 7F001100> LOAD AAHCIA.S INIT 0,00100000 END</pre>
4	If two CLMI boards are used, then add the following directives after the first END: <pre>CLMI 2,6 LOAD <CLMI.S or CCP.S> RUN <41100 or 7F001100> LOAD AAHCIA.S INIT 0,10000000 END</pre>

Restart the CLMI board

The next task will be to restart the CLMI board by cycling the CLMI into service. The CLM Platform and the Applied Automation HCI-A application software are loaded in the CLMI whenever the CLMI is placed in service.

The next step

This concludes the Interface Layer Installation Procedure. Continue with **Section 6** of this manual. The next thing you need to do to complete the installation is build your network definition table.

4.6 Adding another Network Interface

Adding additional applications

Each time you add another network to an existing CLM, you need to modify the CLMI configuration file for that CLM to include the new network. This may be as simple as adding another CLMI board (CLMI or CCP) or adding a port to an existing CLMI.

If you are adding another CLM node, then you will need to add a new CLMI directives file for that CLM and will need directives added for that application.

4. Install AM/CLM Network Interface Layer

4.6. Adding another Network Interface

5. Install APP/CLM Network Interface Layer

5.1 Overview

Introduction

This section discusses how to install the CLM and HCI-A applications when using a **APP/CLM node**.

After installation you will need to create the AM image points to reflect your network and establish the connection between external network points and the AM image points. (Refer to *CLM Platform Implementation Manual*, **Sections 6 and 7**.)

Use Section 4 for instructions on how to install the HCI-A application when using an AM-based CLM node.

5.2 APP/CLM HCI-A Application Pre-Installation

Prior to installation

The APP/CLM platform should be installed prior to installing the HCI-A application. See the APP/CLM Platform Planning, Installation Manual (LK02-200) for instructions on installing the APP/CLM Platform.

5.3 Windows 2003 Setup and Installation



ATTENTION:

This section is only applicable for users who will continue to use the TPS node on Windows 2003 operating system.

Shutdown TDC Emulation Service

In the Windows 2003 environment, the Honeywell TDC Emulator Service simulates the peripheral boards that exist in a traditional TDC 3000 AM/US node chassis. In the APP/CLM, this service emulates the CLMI and CCP boards found in an AM/CLM. The TDC Emulators Service must be stopped before installation of the APP/CLM Application.

Step	Action
1	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.

5. Install APP/CLM Network Interface Layer

5.3. Windows 2003 Setup and Installation

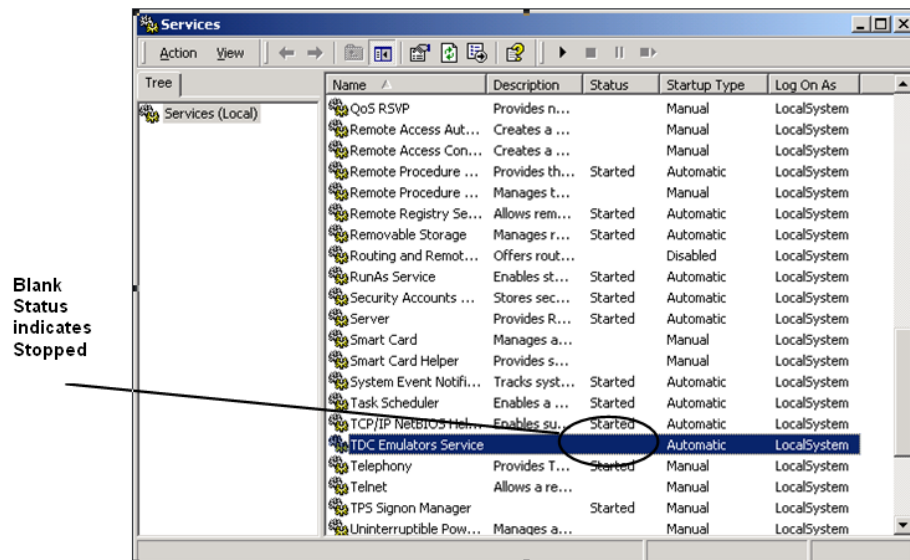
Step	Action
2	From the Start menu select Settings/Control Panel/ Administrative Tools .
3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears.
4	If the Status for TDC Emulators Service is Started , select the TDC Emulators Service entry. From the drop-down menu, select Action , then the Stop button.

A message box appears indicating that the service is being stopped. This will take about 30 seconds.

NOTE: Occasionally, the TDC Emulator Service does not stop in the required time and an Error Message box appears with the message "Could not stop TDC Emulator Service service". When this happens, press the OK button in the message box. The TDC Emulator Service will normally stop within 10 additional seconds. This can be verified by pressing the Ctrl, Alt and Del keys simultaneously and invoking the Task Manager. View the Processes tab from the Task Manager and verify that the *tdcserver.exe* process is no longer in the active process list.

See Figure 5.1 below.

Figure 5-1 TDC Emulators Service Stopped



Install APP/CLM/HCI-A Application

Table 5-1 Install TPS APP/CLM/HCI-A Application

Step	Action
1	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2	Insert the Windows 2003 APP/CLM HCI-A Application distribution CD-ROM into the drive.
3	View the distribution media in the drive using Windows Explorer. From Windows Explorer launch the CLM AAHCIA Install.exe program on the distribution media.
4	The Select CLM Starting COM port window opens up. • Select COM1 if you are using the standard PC ports. • Select COM3 if you are using the COM port Systech Extender that are installed.
5	Follow the instructions for installation. It is recommended that any defaults be selected.

5.4 Windows 2008 Setup and Installation



ATTENTION:

This section is only applicable for users who will be using the Experion node on Windows 2008 operating system (both 32 bit and 64 bit variants).

Shutdown TDC Emulation Service

In the Windows 2008 environment, the Honeywell TDC Emulator Service simulates the peripheral boards that exist in a traditional TDC 3000 AM/US node chassis. In the APP/CLM, this service emulates the CLMI and CCP boards found in an AM/CLM. The TDC Emulators Service must be stopped before installation of the APP/CLM Application.

Step	Action
1	On the Experion APP (Windows side of the APP/CLM), log onto Windows as Administrator.

5. Install APP/CLM Network Interface Layer

5.4. Windows 2008 Setup and Installation

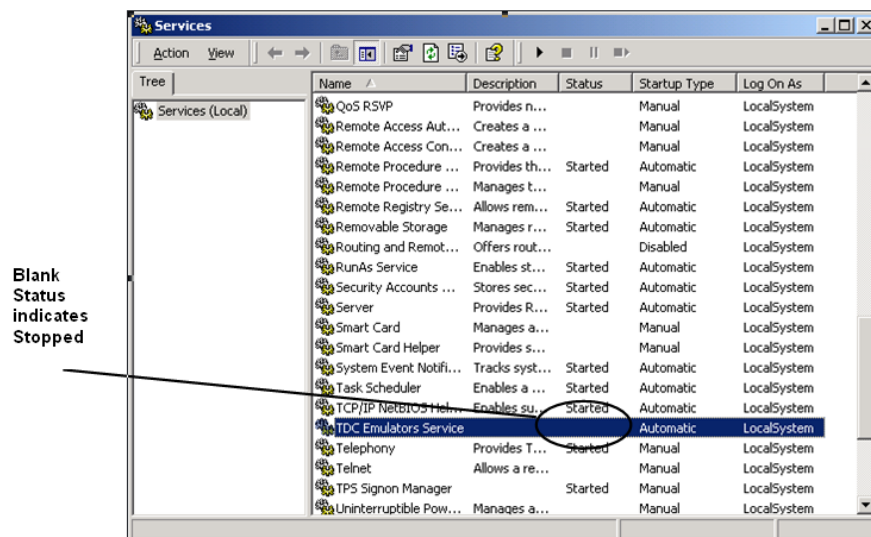
- | Step | Action |
|------|---|
| 2 | From the Start menu select Control Panel /System and Security/Administrative Tools . |
| 3 | Select Services and in the Services window, scroll down until the TDC Emulators Service appears. |
| 4 | If the Status for TDC Emulators Service is Started , select the TDC Emulators Service entry. From the drop-down menu, select Action , then the Stop button. |

A message box appears indicating that the service is being stopped. This will take about 30 seconds.

NOTE: Occasionally, the TDC Emulator Service does not stop in the required time and an Error Message box appears with the message "Could not stop TDC Emulator Service service". When this happens, press the OK button in the message box. The TDC Emulator Service will normally stop within 10 additional seconds. This can be verified by pressing the Ctrl, Alt and Del keys simultaneously and invoking the Task Manager. View the Processes tab from the Task Manager and verify that the *tdcaserver.exe* process is no longer in the active process list.

See the figure below.

Figure 5-2 TDC Emulators Service Stopped



Install APP/CLM/HCI-A Application

Table 5-2 Install Experion APP/CLM/HCI-A Application

Step	Action
1	On the Experion APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2	Insert the Windows 2008 APP/CLM HCI-A Application distribution CD-ROM into the drive.
3	View the distribution media in the drive using Windows Explorer. From Windows Explorer launch the CLM_AAHCIA_Install.msi program on the distribution media.
4	The Select CLM Starting COM port window opens up. • Select COM1 if you are using the standard PC ports. • Select COM3 if you are using the COM port Systech Extender ports that are installed.
5	Click Next and follow the on-screen instructions for installation. It is recommended that any defaults be selected.

5.5 Configure the APP/CLM/HCI-A Application & Port

Table 5-3 Configure the APP/CLM/HCI-A Application & Port

Step	Action
1	From the Start menu on the TPS APP node select Programs/Honeywell TPS and start the Configuration Utility . Note: In Windows 2008, select Programs, Honeywell Experion PKS, System Management, and Configuration Utility from the Start menu.
2	From the Configure menu, select CLM . The CLM Configuration page appears as shown in Figure 5.3.
3	Select the Virtual Board number (1 or 2) and logical chassis slot to be used. The recommended slot is slot 3 (use only slot 2 or slot 3). NOTE: This same slot number must be used in the later AM/CLM configuration. It will be used in the CLMIOnn.ZC file (see Section 5.7).

5. Install APP/CLM Network Interface Layer

5.5. Configure the APP/CLM/HCI-A Application & Port

- 4 Enable the Check Box for one of the Applications and from the list of Applications select **HCI-A**.

Note: An application may be configured only once per CLMI.

- 5 Select the **Ports** to be configured in the CLM Database for this HCI-A application.

NOTE: The same port numbers must be used in the later **CLM NDT** configuration (see **Section 6.2**).

- 6 Select the **OK** Button and exit the Configuration Utility.
-

Figure 5-3 CLM Configuration Page Example

The screenshot shows the CLM Configuration Page with tabs for Virtual Board 1 and Virtual Board 2. The Virtual Board 1 tab is active. The page is divided into sections for CLM Interface, Application 1, Application 2, Application 3, and Application 4. Application 1 is set to AA HCI-A with Port 1 selected. Application 2 is set to Modbus with Port 2 selected. Application 3 is set to OPC with Port 3 selected, and its Port 3 field is populated with 'Matrikon.OPC.Modbus'. Application 4 is currently disabled. The bottom of the window has OK, Cancel, Apply, and Help buttons.

Section	Setting	Port 1	Port 2	Port 3	Port 4
CLM Interface	☒ CLMI, + Warning (EW)				
Application 1	AA HCI-A	☒	☐	☐	☐
Application 2	Modbus	☐	☒	☐	☐
Application 3	OPC			☒	☐
Server ID				Matrikon.OPC.Modbus	
Remote ID					
Application 4		☐	☐	☐	☐

Start TDC Emulator Service

The TDC Emulator Service simulates the CLMI and CCP boards found in the AM/CLM. Before the TPN personality can be loaded the TDC Emulator Service must be started.

Note: With the following exceptions, when the CLM configuration is changed from the CLM page of the TPS Configuration Utility, the TDC Emulator Service must be stopped and re-started before the changes will take effect.

5. Install APP/CLM Network Interface Layer

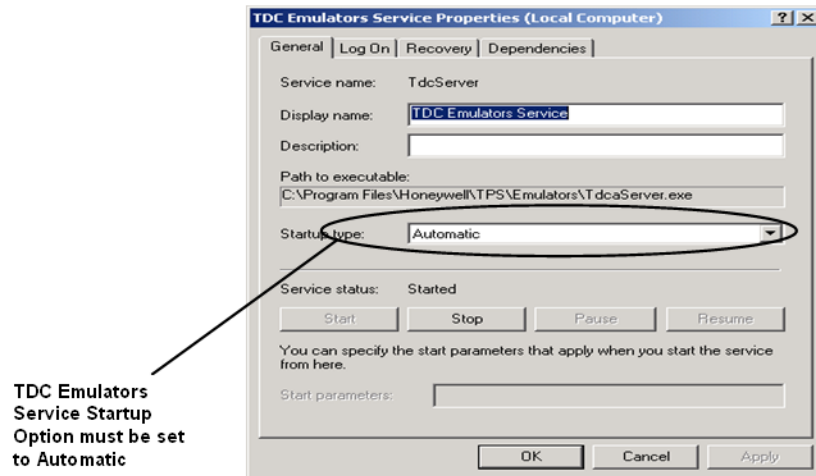
5.5. Configure the APP/CLM/HCI-A Application & Port

Exceptions are when the **Ports** of a previously configured APP/CLM HCI-A Application entry are changed. In these cases, the changes will take effect when the CLMI board is taken Out of Service and then put back In Service.

Table 5-4 Start TDC Emulator Service

Step	Action
1	Log onto Windows as Administrator.
2	From the Start menu, select Settings/Control Panel/ Administrative Tools .
3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears.
4	Select the TDC Emulators Service entry.
5	From the drop-down menu, select Action , then Properties .
6	If the Startup Type is not Automatic, then select the drop-down menu and change it to Automatic . See Figure 5.4 below.
7	Select OK to accept the changes.

Figure 5-4 TDC Emulators Service – Auto S/U



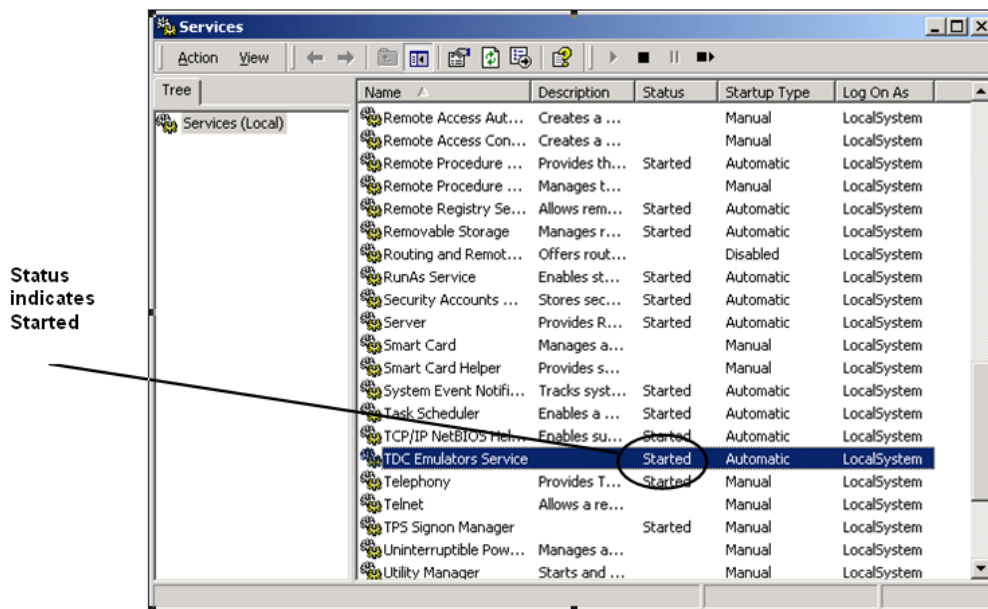
5. Install APP/CLM Network Interface Layer
5.5. Configure the APP/CLM/HCI-A Application & Port

Step	Action
8	If the Status for TDC Emulators Service is Stopped, select the TDC Emulators Service entry. From the drop-down menu, select Action, then the Start button. A message box appears indicating that the service is being started. This will take about 10-15 seconds. The Services display should appear as in figure 5.5 below.
9	At this point the AM Personality may be loaded on the CLM. It is recommended that the APP/CLM node be reset from the LCNP Status Display before loading the APP Personality. To achieve this, from the Start menu select Programs/Honeywell TPS, LCNP Status, select the Detail button and select the Reset button. In Windows 2008 select All Programs, Honeywell Experion PKS, TPS applications, LCNP status, select the Detail button and select the Reset button.

5. Install APP/CLM Network Interface Layer

5.6. Install Network Interface Layer - Setup

Figure 5-5 TDC Emulators Service Started



The next step

This concludes the APP/CLM Windows Interface Layer Installation. Continue with the APP/CLM HCI-A Installation in **Section 5.3**.

5.6 Install Network Interface Layer - Setup

Overview

There are certain steps that must be performed to assure the process is successful. This installation process assumes a working knowledge of the Engineering Personality functions.

Prior to installation procedure

NOTE: The installation of the CLM Platform product should be complete before continuing this installation.

Table 5-5 APP/CLM Pre-Install Instructions

Step	Action
1	Make a copy of the release volume.
2	Verify the path for the CL SOURCE/OBJ. We suggest that you use UCLM.
3	Verify the path for the CL CUSTOM GDF=CDSG. We recommend that only one network directory be used, and not changed.
4	Set the CL PARAM LIST to the same path as the CL SOURCE/OBJ. This information is used when certain CDS modules are compiled.
5	Set the USER DEFLT PATH to the same path as the CL SOURCE/OBJ. This information is used when the supplied CDS modules are compiled and when the executable module link and unlink directives are executed.
6	Review the files in the product release which will be copied to your system to assure no conflicts (duplicate names) in file naming with files already on your system. This is especially critical for CL/CDS files which may be overwritten.
7	Before the install procedure is executed, the paths must be identified and the volume of NET>&CUS> must exist. Normally, this directory is created during the installation of the AM.

Files included in the product release are copied to your system during the install process.

5.7 Copy Install Files to HM

Copy files from distribution media

In the command function, within the engineering personality, execute the following EC and follow all of the prompts given within the EC procedure.

WARNING	Recompilation will require that all points utilizing the CDS to be deleted and rebuilt. Recompilation of an existing CDS will change the date of the object file and this will cause a CDS mismatch to occur between points built with the original CDS and the newly compiled one (see Table 5.7 Step 8).
----------------	--

5. Install APP/CLM Network Interface Layer

5.7. Copy Install Files to HM

Table 5-6 Installation EC Procedure

Step	Prompt or Message	Action
1		The following Executes the AA_INST EC: Enter EC \$Fn>HCIA>AA_INST.EC \$Fn Where n = the Bernoulli device identification (for example, \$F1)
2	A check is made to see if a certain file exists that would make this an upgrade. If this is a reinstallation, a message appears and gives user a chance to quit. DO YOU WISH TO CONTINUE?	Answer Y and the EC will continue. Answer N and the EC will exit.
3	Are your paths acceptable?	Answer Y and the EC will continue. Answer N and the EC will exit.
4	The question what is the destination directory for the CL Source/Object.	Enter NET>UCLM where UCLM is the path name of the destination directory
5	The question what is the destination directory for the custom display files.	Enter NET>UCLM where UCLM is the path name of the destination directory
6	The question what is the destination directory for the DATA BUILD files.	Enter NET>UCLM where UCLM is the path name of the destination directory
7		Copy other required files to NET>&CUS>.

Step	Prompt or Message	Action
8		ZI, ZT, ZS, ZC, AAHCIA.S
	If reinstallation, ask if want to re-compile the CDS.	If not reinstall then compiles CDS.
	Do you want to continue compiles with overwrite switch?	If reinstall and N, then exit. If reinstall and Y, then compile CDS with overwrite switch.
	Note: If any errors exist, they must be corrected before continuing. Then the package will be resident on the destination LCN and installation procedures may continue. The Custom Displays included must be compiled in the desired directory. (Please see Schematics at Section 6.3)	

5.8 Complete CLM Platform

Install Platform

Before proceeding, you must complete the CLM Platform installation if you have not already done so.

Note: You must build the CLM and Builder points in the AM database before you attempt to restart the node with the ZC file described in the next section.

5.9 Configure the CLMI Board & Slot

Communication protocol layer

This section defines the modification of the configuration information required by the platform to identify what hardware and software is used to accomplish communications with the external devices. The CLM Platform provides the lower level functionality to support the protocol layer on top of the platform. The Honeywell developed protocol layer will include the software and other pieces required to communicate with the external devices. This protocol layer is the Applied Automation Optichrom Advance System HCI-A interface.

CLMI load directive file

Each APP/CLM node can have up to two emulated CLMI boards and it must have its own CLMI load directive file. This configuration file is an ASCII text file with the following naming convention:

5. Install APP/CLM Network Interface Layer

5.9. Configure the CLMI Board & Slot

- The name is the literal CLMI followed by three digit CLM physical node number and extension ZC (example: CLMI024.ZC)
- The file is stored in the NET>&CUS> directory
For example, NET>&CUS>CLMI024.ZC is the name for physical node 24.

The CLMI load directive file creation is a part of the CLM Platform installation procedure. Unlike the AM based CLM node this implementation requires **NO** modification of that directive file.

The form of the configuration directive for a APP/CLM environment is as follows:

Table 5-7 CLM: CLMI Load Directive

CLMI <board #>, <slot #> END

CLMI slot numbers

Use the CLM Configurator to determine the CLMI slot number for the APP node.

Edit your configuration file

Review the configuration information to the CLMI configuration file that was installed as part of the CLM Platform software and edit it as described below.

Table 5-8 APP/CLM: Edit Your Configuration File

Step	Action
1	Set the current path name to NET>&CUS.
2	Using the text file editor, call up the text file NET>&CUS>CLMI0nn.ZC, where 0nn is the CLM physical node number. For example, CLMI024.ZC is the CLMI configuration file for CLM node 24.
3	Only The CLMI directive is used. The LOAD, INIT and RUN directives are not used for the APP/CLM. Note: The APP/CLM slot must match the slot # used in the configuration of the APP/CLM Platform on the Windows side as described in Section 5.2.3 step 3 . Use only slot 2 or slot 3. The following example configures a single board in slot 3: CLMI 1,3 END

5. Install APP/CLM Network Interface Layer

5.9. Configure the CLMI Board & Slot

Step	Action
	The following example configures two boards in slot 2 and 3: CLMI 1,2 END CLMI 2,3 END Note: The APP/CLM port is configured from the HCI-A application on the Windows side as described in Section 5.2.3 .

Restart the TDC Emulator Service

The next task will be to restart the TDC Emulator Service so that the configuration changes will be used by the APP/CLM node. Table 5.7-3 describes the procedure to restart the TDC Emulator Service and get the APP/CLM back on the operating condition as an LCN node.

Table 5-9 APP/CLM: Restart the TDC Emulator Service

Step	Action
1	Stop the TDC Emulators Service .
2	Start the TDC Emulators Service .
3	The CLM should to into the OK/PWR_ON condition. Reload the CLM from a GUS or US station.

The next step

This concludes the Interface Layer Installation Procedure. Continue with Section 6 of this manual. The next thing you need to do to complete the installation is build your network definition table.

5. Install APP/CLM Network Interface Layer

5.10. Implementing another APP/CLM Port

5.10 Implementing another APP/CLM Port

Adding additional applications & ports

The APP/CLM supports two emulated CLMI boards. A maximum of four ports per board may be configured. Different protocol applications can run on the APP/CLM as long as they are configured for separate ports. A protocol application may be configured only once per CLMI.

Each time you implement a new CLM port you need to configure the port via the CLM configurator and then restart the TDC Emulator Service.

6. Build the Network Interface

6.1 Overview

Introduction

The Interface Design consists of the following tasks.

- Network Design and Configuration
- Network Status and Performance Displays
- Organization of PITs
- The following provides a detailed explanation of each of these tasks.

Addressable components

The CLMI has an eight component address scheme. The Applied Automation Optichrom Advance System network interface uses four addressable components as shown below.

Table 6-1 Addressable Components

Component	Description
NDT Address Components	
CLMI	CLMI Board; 1 or 2 Boards per CLM.
PORT	CLMI RS-232 Port interface; 1 to 4 Ports per CLMI
DEVICE	Analyzer device address (1 - 254)
Address entered in PITE	
POINT	Stream number (1 - 16)

NDT Entries

NDT entries belong to a hierarchy, and must be built in the following sequence:

```
CLMI
  |- PORT
    |- DEVICE
      |- PIT
        |- PITE
```

6. Build the Network Interface

6.2. Network (NDT) Design and Configuration

Note: the PIT entries require the use of the name of a device NDT.

All of the above entries are built before you build the PITE entries for individual points (discussed in Section 7).

6.2 Network (NDT) Design and Configuration

CLM configuration

The following is an outline of the steps used in the design and configuration of a CLM interface.

Obtain and understand the following documents.

- *CLM Platform Implementation Manual*
- Interface Application Manuals (Refer to **Section 7**).
- An overview diagram of your network installation: Identify the address of each analyzer device in your network.
- A listing of the analyzer stream and component data to be imaged by LCN points in your installation.

Table 6-2 CLM Configuration

Step	Action								
1	Identify the Template Definition Files (TDFs) for each type of addressable component in network. The TDFs are included by Honeywell and will be used in building the Network configuration. The CLMI TDF is detailed in the <i>CLM Platform Implementation Manual</i>. Specific NDT TDFs are: <table><tr><th>Template Name (ZT)</th><th>Description</th></tr><tr><td>CLMI</td><td>CLMI Interface Board (Platform)</td></tr><tr><td>AA_PORT</td><td>CLMI Port</td></tr><tr><td>AA_DEV</td><td>Network Device / Analyzer</td></tr></table>	Template Name (ZT)	Description	CLMI	CLMI Interface Board (Platform)	AA_PORT	CLMI Port	AA_DEV	Network Device / Analyzer
Template Name (ZT)	Description								
CLMI	CLMI Interface Board (Platform)								
AA_PORT	CLMI Port								
AA_DEV	Network Device / Analyzer								
2	The TDF and STSDF files were transferred to the HM directory NET>&CUS in the installation process previously accomplished.								

Step	Action
------	--------

- 3** Identify each addressable network component.

Component	Description	Range
CLMI	Communications Link Module Interface Board	1-2
PORT	Communication Port on the CLMI	1-4
DEVICE	Analyzer Device Address Note: This is the address of the analyzer within the Network that is attached to the Port identified above.	1-254

- 4** **Step 4 and the creation of a BIF is optional. You can configure the network on-line using the interactive build feature of the CLM Configuration Builder.**

Create a BIF to include one NDT entry for each of the identified addressable network components. The BIF may be created using the TPS file editor, or other text file editor, and transferred to the HM directory of your choice (default is NET>&CUS).

Names of the prototype BIFs included by Honeywell are listed in Table 3.4).

Typical BIF listings for the CLMI, Port and Device are included in **Section 6.2.**

Copy the appropriate BIF file, and then complete each definition within the BIF, to create the BIF NDT entry for each addressable network component in your network.

- 5** Using the CLM Configuration Builder, build the NDT entries on the target CLM node.

Instructions for using the CLM Configuration Builder are detailed in the *CLM Platform Implementation Manual*.

6. Build the Network Interface

6.3. Port NDT Builder Input File Format

CLMI NDT Builder Input File Format

Introduction

The following Network Definition Table (NDT) Builder Input File Formats are required for this interface. These forms are included as BIF's with the extension .ZI.

Table 6-3 NDT BIF's

Template Name (.ZI)	Description
CLMI	CLMI Interface Board (Platform)
AA_PORT	CLMI Port
AA_DEV	CLM Analyzer Device

CLMI Builder Input File Format

The BIF shown in Table 6.3 is for the Communication Link Module Interface (CLMI). Build one CLMI entry for each CLMI/CCP board in the CLM. The user must complete the form by supplying the name and address data. Refer to Table 6.4 for an explanation of CLMI BIF entries.

Table 6-4 CLMI/Typical BIF for the CLM Interface

```
ENTRY   = CLMI
NAME    =
A1_CLMI =
FAILOVER =
END
```

Identifying CLMI boards

Step	Field	Action
1	ENTRY	CLMI
2	NAME	Enter a name identifier for the CLMI board component. The suggested names are CLMI_1 for the first CLMI board and CLMI_2 for the second CLMI board.
3	A1_CLMI	Enter the CLMI number: 1 or 2. Enter 1 for the first CLMI board and 2 for the second CLMI board.

Step	Field	Action
4	FAILOVER	(Y/N) Enter N.

6.3 Port NDT Builder Input File Format

AA_PORT Builder Input File Format

The BIF for the ports is shown. Build one AA_PORT entry for each port connected to the network. Refer to Table 6.5 for an explanation of AA_PORT BIF entries.

NOTE: Modem Control is not supported with this protocol template.

Table 6-5 AA_PORT/Typical BIF for the Network

```
ENTRY  = AA_PORT
NAME   =
A1_CLMI =
A2_PORT =
* Baud (0=300 1=1200 2=2400 3=4800 4=9600 5=19200)
BAUD   = 4
DBITS  = 7
SBITS  = 2
PARITY  = 2    * (NONE=0,ODD=1,EVEN=2)
RETRY   = 2
TIMEOUT = 5
END
```

Identifying CLMI Ports

Table 6-6 BIF Fields Identifying CLMI Ports

Step	Action	Result
1	ENTRY	AA_PORT -- Template Name

6. Build the Network Interface

6.3. Port NDT Builder Input File Format

Step	Action	Result
2	NAME	Enter up to an 8 character name identifier for the port component. The suggested names are PORT_1 for the first port, PORT_2 for the second, etc.
3	A1_CLMI	Enter the CLMI number that the port is on: 1 or 2.
4	A2_PORT	Enter the port number: 1 through 4
5	BAUD	Enter the baud rate code for this communication port. Valid codes and their meanings are:

Code	Baud Rate
0	300
1	1,200
2	2,400
3	4,800
4	9,600 (Default)
5	19,200

6	DBITS	Enter the number of data bits, (Default 7)
7	SBITS	Enter the number of stop bits, 1 or 2 (Default 2)
8	PARITY	Enter the parity code. Valid codes and their meanings are:

Code	Parity
0	None
1	Odd
2	Even (Default)

Step	Action	Result
9	RETRY	Enter the RETRY code. This defines the number of retries that will be attempted on a communication failure before aborting the request. The default is 2, which means a total of 3 attempts to communicate will be made before aborting the request.
10	TIMEOUT	Request time-out period. This is the base time-out period (in seconds) for an OK, or NO response from the HCIA before the request is retried. (Default 5.)

AA_DEV Builder Input File Format

The BIF shown is for the device. Build an AA_DEV entry for each analyzer device in the network. Refer to for an explanation of AA_DEV BIF entries.

Table 6-7 AA_DEV/Typical BIF for the Analyzer Device

```
ENTRY  = AA_DEV
NAME   =
A1_CLMI =
A2_PORT =
A3_DEV1 =
RPT_TIME =
END
```

6. Build the Network Interface

6.3. Port NDT Builder Input File Format

Identifying Device

Table 6-8 Identifying Analyzer Devices

Step	Action	Result
1	ENTRY	AA_DEV -- Template Name
2	NAME	Enter a name identifier for the analyzer device. The name must be unique in the CLM node.
3	A1_CLMI	Enter the CLMI number: 1 or 2.
4	A2_PORT	Enter the port number: 1 to 4.
5	A3_DEV1	Enter the address of the analyzer device which has a range of 1 to 254. This is the address of this device within the Network.
6	RPT_TIME	This is a time, in minutes, within which at least one stream of this analyzer will send a results message. If no result is received and this time is elapsed, and a result message not received for any of the streams on the device, then the device is FAILED. Default 5

TDFs for PIT templates

The TDFs for the PIT templates are common across all CLM implementations.

Instructions and listings for the PIT Template Definition Files are detailed in the CLM Platform Implementation Manual. Refer to it for additional information.

The PIT essentially creates a point processing table for a specific device. When PITE's are built for image points, the image point is assigned to a specific PIT, which specifies the CLMI, Port, and Device address for the PITE.

CLMI Builder Input File Format

Build one PIT entry for each device (analyzer).

NOTE: The HCI-A PITs should **NOT** be scheduled. Scheduling the PIT's will cause unnecessary loading, since all analyzer values are unsolicited input. Since PITs are not scheduled, only one PIT is required for each device.

Table 6-9 PIT BIF for the CLM Interface

```
ENTRY  = PIT
NAME   =
CYCLE  =
CYC_ALM=
SLO_DWN=
NDT_NAME =
GRP_NAME =
END
```

Step	Action	Result
1	ENTRY	PIT Defines the template used to the device.
2	NAME	Enter the associated device name. Refer to the AA_DEV files created in Section 6.2.3 .
3	CYCLE	-1 Enter the cyclic rate at which the PIT is to be processed. Minus 1 means that the PIT is unscheduled. All HCI-A PITs should be set to a cycle of -1. N
4	CYC_ALM	This entry is an alarm notification option, used to force an alarm message if the specific PIT processing falls behind the configured schedule.(0-256). No default. Not used for HCI-A. Set to 10.
5	SLO_DWN	This value defines the number of normal cycles to process before processing parameters defined as slow parameters. Default is 5. Not used for HCI-A.

6. Build the Network Interface

6.3. Port NDT Builder Input File Format

Step	Action	Result
6	NDT_NAME	Name of the device NDT this PIT is to be assigned to. NOTE: Since all HCI-A input is unsolicited, and the PITs are unscheduled, only one PIT should be built per device.
7	GRP_NAME	Optional entry. Name of an associated PIT Group. If entered, this PIT will be a member of the named PITG.

NDT Status Displays

Introduction

The following figures are for information only. These figures define the status display layouts for the CLM, Port, and Device level files in the application

Table 6-10 Status Display for CLMI

###	– NO. OF UNACKNOWLEDGED FAILED DEVICES ON THE CLMI	
###	– NO. OF ACKNOWLEDGED FAILED DEVICES ON THE CLMI	
Average Parameters per Second		
	INPUT	OUTPUT
4 Sec.	####	####
5 Min.	####	####
##### – NO. OF PPSs [CLM_PPS, CLM_OUT and PPS]		

Table 6-11 Status Display for PORT

Applied Automation HCI-A Release 1 Revision 1	
### – NO. OF UNACKNOWLEDGED FAILED DEVICES ON THE PORT	
### – NO. OF ACKNOWLEDGED FAILED DEVICES ON THE PORT	
##### – MESSAGES TRANSMITTED	##### – MESSAGES REJECTED
##### – MESSAGES RECEIVED	##### – CHECKSUM ERRORS
##### – TIMEOUTS	##### – PROTOCOL ERRORS
Start Char - STX	End Char - NONE
Baud Rate ##### Retries ### Timeout ### Data Bits 7 Parity Even	

Table 6-12 Status Display for Analyzer Device

##### – MESSAGES TRANSMITTED	##### – MESSAGES REJECTED
##### – MESSAGES RECEIVED	##### – CHECKSUM ERRORS
##### – TIMEOUTS	##### – PROTOCOL ERRORS
##### – REPORT TIMEOUT IN MINUTES	

6. Build the Network Interface

6.3. Port NDT Builder Input File Format

7. Build Data Base Points

7.1 Image Point Design and Configure

Image point design and configuration process

The following Table is an outline of the steps used in the design and configuration of an image point interface to a network device. Obtain and understand the following documents:

- CLM Platform Implementation Manual
- User's Manuals for your remote devices.

Note: Before designing and configuring image points, you must have completed Network Design and Configuration. This activity is described in Section 6.

Step	Action
1	Identify the analyzer streams from the remote network that are to be reflected in AM image points.
2	Define the AM image points.
3	Identify the network structure for each AM image point. This includes board structures, port structures, device structures, PIT structures, and PITE structures. The TDFs for the PIT are listed in the <i>CLM Platform Implementation Manual</i> . The TDF files are a part of the product and can be looked at on your system.
4	The TDF and image point files were transferred to the HM directory NET>&CUS in the installation process previously accomplished.
5	Identify the specific points in the network that require AM image points.
6	Get the data address information.
7	Step 7 and the creation of a BIF is optional. You can configure the network on-line using the interactive build feature of the CLM Configuration Builder. Add data to a BIF for each of the identified image points. Actual BIF listings for each point type are included in Section 7.4 of this manual. Copy the appropriate BIF file and complete each definition within the BIF to create the BIF PITE entry.

7. Build Data Base Points

7.2. AM Points

Step	Action
	The BIF may be edited using the LCN file editor, or other text editor.
	E Each PITE entry must be input to the CLM Configuration Builder to generate the PITE interface structure required for each imaged point.
8	Build the image points on the target CLM node. This is performed using the LCN Data Entity Builder. Instructions for using the LCN Data Entity Builder are detailed in the <i>LCN Data Entity Builder Manual</i> .
9	Build the point interface structures on the target CLM node. This is performed using the CLM Configuration Builder, which is installed on the CLM node as part of the CLM platform software. Instructions for using the CLM Configuration Builder are detailed in the CLM Platform Implementation Manual.

7.2 AM Points

Image point composition

In this subsection, each interface point configuration, CDS and CL is defined. A subsection is also included that outlines the creation of the points.

The interface point are:

- One REGDUM point
- One CLM Analyzer Stream Status Point
- Stream points, as many as required
- Component points, as many as required

All the image points should be built on the AM prior to building the PITEs. The image point must exist on the LCN system prior to building the respective PITE.

Dummy Regulatory Point

Introduction

The Dummy Regulatory Point is created on the system. This point is used by the stream point to define an array type. Refer to the REGDUM.EB file supplied with this interface. The following defines the point configuration, CDS and CL.

Dummy Regulatory Point Configuration

Configuration parameters for the REGDUM point. Modify the REGDUM.EB file accordingly.

Table 7-1 Dummy Point Configuration

Parameter	Description	Entry/Notes
&N	POINT NAME	REGDUMMY <u>Do not change name!</u>
UNIT	UNIT ID	Enter Unit - Two characters. Required user entry!
PTDESC	POINT DESCRIPTOR	"DUMMY STREAM POINT "
EUDESC	EU DESCRIPTION.	" "
KEYWORD	KEYWORD DESCRIPTION.	" "
PRIMMOD	TAG NAME OF POINT.	--
PTDISCL	SELECT THE FULL DISCLOSURE OF ALL APPLICABLE PARAMETERS IN THE PED, OR A SUBSET.	BRIEF
PVALGID	PV ALGORITHM IDENTIFIER.	= NULL
CTLALGID	CONTROL ALGORITHM USED BY THIS POINT.	NULL
OVERVAL	OVERVALUE INDEX.	25
SUPPIO	PERMITS OF SUPPRESSES DISPLAYS OF THE PV AND OP ON OVERVIEW AND GROUP DISPLAYS.	NOSUPPR

7. Build Data Base Points

7.2. AM Points

Parameter	Description	Entry/Notes
\$IPPASN		OFF
PERIOD	THE INTERVAL THIS POINT IS SCHEDULED FOR PROCESSING.	NOPERIOD
ALENBST	POINT ALARM-ENABLE STATUS THAT AFFECTS THE DETECTING AND REPORTING OF ALARMS.	ENABLE
ALPRIOR	POINT ALARM PRIORITY.	LOW

Dummy Regulatory Point CDS parameters

This point does not require any CDS packages.

Dummy Regulatory Point CL parameters

This point does not require any CL.

Dummy Regulatory Point Build

From the Engineering Personality Main Menu, select the Builder Commands target.
From the Command Display select Exception Build and build REGDUM.EB

CLM Analyzer Stream Status Point

The CLM Analyzer Stream Status Point (OPTANLZ) is used by schematic CLMANLZA to access all the stream points. The following subparagraphs define the point configuration, CDS and CL.

CLM Analyzer Stream Status Point configuration

The stream overview point is an AM Custom Data Point Configuration (AM88-350).
Modify the OPTANLZ.EB file for your system.

Table 7-2 CLM Analyzer Stream Status Point Configuration

Parameter	Description	Entry/Note
&N	POINT NAME	OPTANLZ Do not change name!
UNIT	UNIT ID	Enter Unit Two characters. Required user entry!
PTDESC	POINT DESCRIPTOR	"STREAM OVERVIEW DISPLAY "
KEYWORD	POINT KEYWORD	
CLSLOTS	NUMBER OF CL BLOCKS	0
NOPKG	No. OF CUSTOM DATA PACKAGES	1
PRIMMOD	PRIMARY MODULE POINT ID	--
ALPRIOR	POINT ALARM PRIORITY	Low
ALENBST	POINT ALARM ENABLE STATUS	Inhibit
PKGNAME(1)	SOURCE FILE 1 NAME	ANLZMAIN
PERIOD	POINT EXECUTION PERIOD	NOPERIOD

CLM Analyzer Stream Status Point CDS parameters

CDS parameters come from one package and appear on the Custom Page. Refer to Table 7.3 for the parameters in the ANLZMAIN CDS package.

7. Build Data Base Points

7.2. AM Points

Table 7-3 CLM Analyzer Stream Status Point CDS Parameters

Parameter	Description	Notes
STRMPNTS	Stream Point ID Array (1..200)	Required User Entry from the CLMANLZA schematic
STRMCONF	Stream Configured Flag Array (1..200)	1.0 = Configured

CLM Analyzer Stream Status Point CL

The CLM Analyzer Stream Status point provides data to the CLMANLZA schematic, and does not require any CL blocks.

CLM Analyzer Stream Status Point Build

From the Engineering Personality Main Menu, select the Builder Commands target.
From the Command Display select Exception Build and build OPTANLZ.EB

Stream Point (AM Switch Point)

The Stream Point images a single stream from an Applied Automation Optichrom Advance System analyzer. The following defines the point configuration, CDS and CL. You may find an included sample EB file helpful. It is AAPNTS.EB and should reside in the directory of NET>UCLM.

Stream Point configuration

The stream point is an AM Switch Data Point Configuration (AM88-350).

7. Build Data Base Points

7.2. AM Points

Table 7-4 Stream Point Configuration

Parameter	Description	Entry/Note
&N	POINT NAME	Required user entry.
UNIT	UNIT ID	Unit – 2 characters Required user entry!
PTDESC	POINT DESCRIPTOR	Required user entry.
KEYWORD	POINT KEYWORD	
CLSLOTS	NUMBER OF CL BLOCKS	1
NOPKG	No. OF CUSTOM DATA PACKAGES	2
NUMSWTCH	NUMBER OF SWITCHES	1
PRIMMOD	PRIMARY MODULE POINT ID	--
SALMDSC1	ALARM 1 DESCRIPTION	
SALMDSC2	ALARM 2 DESCRIPTION	
SALMDSC3	ALARM 3 DESCRIPTION	
ALPRIOR	POINT ALARM PRIORITY	NOACTION JOURNAL LOW HIGH
ALENBST	POINT ALARM ENABLE STATUS	Enable Disable Inhibit
PKGNAME(1)	SOURCE FILE 1 NAME	HCIASTRM
PKGNAME(2)	SOURCE FILE 2 NAME	CLM_DMD3
S1NSTATE	NUMBER OF SWITCH STATES	5
S1STATES(0)	#1 STATE NAME # 1	OFFLINE

7. Build Data Base Points

7.2. AM Points

Parameter	Description	Entry/Note
S1STATES(1)	#1 STATE NAME # 2	NO_ERROR
S1STATES(2)	#1 STATE NAME # 3	WARNING
S1STATES(3)	#1 STATE NAME # 4	LIMIT
S1STATES(4)	#1 STATE NAME # 5	INVALID
S1ACCLVL	ACCESS LEVEL	Operator Supervis Engineer Program
S1CURSTS	INITIAL STATE	INVALID
S1REQSTS	REQUESTED STATE	NO_ERROR
PERIOD	POINT EXECUTION PERIOD	24HR
ALMOPT	ALARM OPTION (on custom page)	(build visible) 0=none, 1=off normal 2=change of state
PVNORMAL	PV VALUE'S NORMAL STATE TEXT	\$ORD0 \$ORD1 Build visible Last two states should not be PVNORMAL

Stream Point CDS parameters

CDS parameters come from two packages and appear on the Custom Page.

HCIASSTRM - with this package (see Table 7.5)

CLM_DMD3 - CLMI Platform (see Table 7.6)

NOTE: The timeout parameter for the stream point should be one greater than twice the cycle time for the stream, or sixty-one seconds, whichever is greater. The period of the stream point should be set with 24HR. The cycle time specifies the period for the stream data to be received from the analyzer.

7. Build Data Base Points

7.2. AM Points

Table 7-5 HClASTRM Stream Point CDS Parameters

Parameter	Description	Notes
ANALYZER	Analyzer Number	Read from stream.
STREAM	Stream Number Control	Read from stream.
NUMSTMCM	No. of Components Expected	Required User Entry. This number must match the number of components specified in the data stream.
NUMCOMP	No. of Components from Stream	Read from stream.
ANASTS	Analyzer Status	Read from stream.
ANAMODE	Analyzer Mode	Not used.
ALRMCODE	Alarm Code	Read from stream.
TIMEOUT	Time in seconds to delay before alarming when sample not received s/b set at (the largest of) twice the (CYTIME * NUMCOMP) + 1 or 61 seconds. -1 no alarm	Required User Entry
SPTIME	Sample Time	Read from stream.
SPTIMELS	Last Sample Time	Set by CL code.
CYTIME	Max. No. Seconds between Samples. Not the same as TIMEOUT	Required User Entry
COMP	Component Stream Value (Array 1..40)	Read from stream.
COMPPT	Component Point ID (Array 0..40) where COMPPT(0) is specified	First entry must be REGDUMMY. The CDS already defaults to this value.

7. Build Data Base Points

7.2. AM Points

Parameter	Description	Notes
COMPCONF	Component Point Configured (Array 1..40)	1.0 = CONFIGURED. 0.0 = NOT CONFIGURED.
COMMAND	Last Command Sent	From Graphic.
COMMVAL	Command to Send	From graphic.
COMMTXT	Text for Send Command	From graphic.
STRMDISP	Stream Display Name	Must be set to: CLMHCI_B
SPTIME_N	Next Sample Time Seconds	Set by CL code.
SPTIME_L	Last Sample Time Seconds	Set by CL code.
SPTIME_C	Current Sample Time Seconds	Set by CL code.
SPTIMH_L	Last Sample Time Hour	Set by CL code.
SPTIMM_L	Next Sample Time Minutes	Set by CL code.
SPTIMS_L	Last Sample Time Seconds	Set by CL code.
SPTIMH_N	Next Sample Time Hour	Set by CL code.
SPTIMM_N	Last Sample Time Minutes	Set by CL code.
SPTIMS_N	Next Sample Time Seconds	Set by CL code.
TIMEOUTO	Track TIMEOUT Change	Set by CL code.
CYTIMEO	Cycle Time Old	Set by CL code.

Note: This Interface supports the HCI-A protocol Host Activation of an event (HAE) message. It can be commanded from the Stream Component Status Display Command Target. To configure this option you must set parameter values as follows:

If COMMVAL <> -1 then show the target

If COMMVAL <> 0 then follow prompts to send Command set up

If COMMVAL = 0 then follow prompts to enter Command to send

Parameter	Description	Notes
CDS Parameters (CLM_DMD3) on Custom Page		
CLM_PPS	STORE TO PROCESS POINT PITE	
CLM_OUT	STORE TO PROCESS OUTPUT PITE	
DEV_STA	DEVICE IN_SERV (ON/OFF)	TRUE = device INSERVICE
PT_FAIL	DEVICE COMPONENT FAILURE (ON/OFF)	TRUE = device component FAIL
HDW_REF	HARDWARE REFERENCE	Not used for this interface.
CLM_PORT	NAME OF PORT NDT	
CLM_DEV	NAME OF DEVICE NDT	
CLM_PIT	NAME OF PIT TABLE	

Stream Point CL

The stream point operates with one CL block - HCI_A2CL.AO.

Theory of Operation

One CL block, at the general insertion point, validates the configuration of ALMOPT, and determines S2CURSTS from the input states. It then performs off-normal alarming when required by the configuration.

Table 7-6 Stream Point CL Blocks

Block Name	Insertion Point	Insertion Order	Description
HCI_A2CL	GENERAL	10	Validates configuration and determines off normal alarms

7. Build Data Base Points

7.2. AM Points

Stream Point Build

From the Engineering Personality Main Menu, select the Application Module target. Select the Switch AM target to build a switch point. Use to define the various entries for that point.x`

Attach CL

Set path for CL SOURCE/OBJ to UCLM or the directory where you installed the CL. Set the CL CUSTOM GDF to CDSG or your choice. Make sure that the stream points are inactive, then link HCI_A2CL to each stream point that was created. Use the following command:

LK HCI_A2CL streampoint name

Build Stream Point PITE

Build an AA_STRM PITE entry for the CLM image point. The BIF entries for the AA_STRM PITE is shown in Table 7.7 and the entry descriptions are listed in Table 7.8.

PITE template assumptions

The PITE makes the following assumptions:

The point is scheduled for alarming of time outs. The AM image point is processed after input by always setting the point option PPS to yes.

Table 7-7 AA_STRM; PITE BIF for the Stream Point

```
ENTRY  = AA_STRM
NAME    =
PIT NAME =
STRM_NUM =
END
```

Table 7-8 PITE Builder Input File Format

Item	Field	Action
1	ENTRY	AA_STRM
2	NAME	Enter the stream point name.
3	PIT_NAME	Enter the associated PIT name.

Item	Field	Action
4	STRM_NUM	Enter the associated stream number.

Stream Component (AM Regulatory Point)

The Stream Component point images a single stream component from the stream point. The stream Component Point is an AM Regulatory Point Configuration (AM88-301). The following defines the point configuration, CDS and CL.

Stream Component Configuration

The purpose of a component point is to get a single component result from the stream and hold it in a Regulatory point that allows alarming.

Table 7-9 Stream Component Configuration

Parameter	Description	Entry/Notes
&N	POINT NAME	Required User Entry
UNIT	UNIT ID	Required User Entry
PTDESC	POINT DESCRIPTOR	Required User Entry
EUDESC	E.U. DESCRIPTOR	Required User Entry
KEYWORD	POINT KEYWORD	Required User Entry
PRIMMOD	PRIMARY MODULE POINT ID	----
PTDISCL	POINT DISCLOSURE	FULL
PVALGID	PV ALGORITHM	DATAACQ
CTLALGID	CONTROL ALGORITHM	NULL
OVERVAL	OVERVIEW DISPLAY VALUE	
SUPPIO	I/O DISPLAY SUPPRESSION	OUTSUPPR
PERIOD	POINT EXECUTION PERIOD	NOPERIOD

7. Build Data Base Points

7.2. AM Points

Parameter	Description	Entry/Notes
Refer to Form "Regulatory PV Algorithms Common Display, AM88-302"		
AM Regulatory Data Acquisition Algorithm (AM88-311)		
NOPINPTS	NO. OF INPUT CONNECTIONS	1
PISRC1	SOURCE POINT ID.PARAMETER 1	STREAM_POINT. COMP(x) ;where x is the array index on the Stream point from which the component point will receive its data.
PIDSTN1	DESTINATION PARAMETER	P1
PIACTSTS1	DEST CONN ACTIVITY STATUS 1	ACTIVE
Refer to Form "Regulatory Alarming and Distribution, AM88-370"		
General Input/Output Connections and Custom Data Segments (AM88-380)		
CLSLOTS	NUMBER OF CL BLOCKS	0
NOPKG	NO. OF CUSTOM DATA PACKAGES	0
NUMSWTCH	NUMBER OF SWITCHES	0
NOGINPTS	NUMBER OF GENERAL INPUTS	0
NOGOPTS	NUMBER OF GENERAL OUTPUTS	0

PISRC1

Specify the parameter that is copied to this point's PV. The parameter is identified by giving the stream point name, and the component entry within the COMP array that gets the component data.

Stream Component CDS parameters

This interface does not require a CDS package.

Stream Component CL

No CL is required for the stream component point.

Stream Component Build

From the Engineering Personality Main Menu, select the Application Module target. Select the Regulatory target to build a regulatory point. Use to define the various entries for that point.

7. Build Data Base Points

7.2. AM Points

8. Build Schematics

8.1 Introduction

Two schematics are supported by the HCI-A application:

CLM Analyzer Stream Status Display

CLM Analyzer Stream Component Status Display

The operator selects schematic CLMANLZA to invoke the Stream Status Display. The operator then may select a stream point to invoke the Stream Component Status Display.

8.2 Compile the Schematics

From the Engineering Personality Main Menu select the Picture Editor and compile schematics which will be installed on the NET>UCLM directory.

CLMANLZA

CLMHCI_B

CLMHCIB2

CLMHCIB3

Table 8-1 Compile Pictures

Step	Action
1	Set the USER DEFLT PATH to the same path as your destination path for displays.
2	The custom display for the application needs to be compiled. To accomplish this, the picture in the following list needs to be read and compiled. CLMANLZA.DS CLMHCI_B.DS
3	In the Engineering Personality, invoke the Picture Editor.
4	Enter the command: R CLMANLZA; COM

8. Build Schematics

8.2. Compile the Schematics

5	Enter the command: R CLMHCI_B; COM
6	Enter the command: R CLMHCIB2; COM
7	Enter the command: R CLMHCIB3; COM

9. Schematic Operation

9.1 Stream Status Displays

Stream Status Displays, which are compiled during the implementation, are two level displays that show the CLM Analyzer Stream Status at the first level, invoked by using SCHEM CLMANLZA or using an assigned button, and the Stream Components Status at the second level, invoked by selecting the desired stream on the entry display/schematic.

The component points must be set ACTIVE in order for the display to show the stream point's data .

CLM Analyzer Stream Status

The CLM Analyzer Stream Status display provides an overview status of up to 200 stream points. Each stream point may image data from different analyzers. Figure 9.1 depicts the CLM Analyzer Stream Status display.

9. Schematic Operation

9.1. Stream Status Displays

Figure 9-1 CLM Analyzer Stream Status Display

07 Nov 94

CLMANLZA

CLM ANALYZER STREAM

STREAM	STREAM	STREAM	STREAM
* A01STR01	* A01STR02	* A01STR03	* A01STR04
* A01STR05	*	*	*
*	*	*	*
*	*	*	*
* A02STR01	*	*	* A02STR04
*	*	*	*
*	*	*	*
*	*	*	*
*	*	*	*
*	*	*	*
*	*	*	*
*	*	*	*
*	*	*	*
*	*	*	*
*	*	*	*

PAGE
FWD

CHANGE
ENTITY

Legend: OFFLINE
NO_ERROR
WARNING
LIMIT
INVALID

Stream Status

The display is color coded to show the status of each stream.

Table 9-1 Stream Status Color Codes

Color Code	Status	Description
White	OFFLINE	Data has not been received from the analyzer stream within the Timeout period, or the analyzer is out of service.
Cyan	NO_ERROR	All components for the analyzer stream are valid.
Blue Full	WARNING	One or more analyzer stream components have a WARNING status or a TIMEOUT.
Yellow	LIMIT	One or more analyzer stream components are out of their configured range.
Red	INVALID	The analyzer stream has an INVALID status.

Adding a Stream Point

The Station must be in Engineering level. To add a new stream point to this display, select an empty slot and then enter the name of the new stream point (existing AM Switch point).

Deleting a Stream Point

The Station must be in Engineering level. To delete a stream point from the display, select the CHANGE ENTITY target and then select the point to be deleted. The point will then be cleared from the display.

Invoking Second Level Display

To invoke the Stream Component Status display, select the stream point name.

Initial Target

The initial target of the display contains the name of the CLM Analyzer Stream Status point which is built during interface installation.

9. Schematic Operation

9.1. Stream Status Displays

Stream Component Status

The Stream Component Status display shows the details of a stream with up to 40 components

Figure 9-2 Stream Component Status Display

07 Nov 94 15:14:22

CLMHCI B

CLM/AA HCI A COMPONENT STATUS

DESC: ANALYZER 1 - STREAM

CYCLE : 00:15

POINT: A01STR01

DETECTOR : 5

STREAM :

LAST RESUT : 10:30

DETECTOR STATE :

STATUS :

NEXT UPDATE : 10:30

ALM CODE :

COMPONENT	DESCRIPTION	EU	PVLOTP	PVHITP
12.500000	COMPONENT 1	PERCENT	10.000000	15.000000
0.0200000	COMPONENT 2	PERCENT	0.0010000	0.0500000

SEND TIME
AND DATE

COMMAND
2ND-STD

DISPLAY
BACK

Component Status

The component status is indicated by the color of the component value.

Table 9-2 Component Status Color for R400

Color Code	Status	Description
Cyan	No Alarm	Component is not in alarm
Red	Emergency Alarm	Component is in emergency alarm condition
Yellow	High Alarm	Component is in high level alarm condition
Full Blue	Low Alarm	Component is in low level alarm condition

Table 9-3 Component Status Color for R500/R600

Color Code	Status	Description
Cyan	NO ALARM	Component is not in alarm
Color not Reversed	ALARM	Component in alarm "Emergency" = Magenta Component in alarm "High" = Red Component in alarm "Low" = Yellow

PV Limit Alarm Indication

The values for PVLOTP and PVHITP are color coded to indicate limit violations.

Table 9-4 PV Limit Color Codes for R400

Color Code	Status	Description
Yellow	No Alarm	PV does not exceed limit (PVLOTP or PVHITP)
Reverse Yellow	Alarm	PV exceeds limit

Table 9-5 PV Limit Color Codes for R500/R600

Color Code	Status	Description
Yellow	NO ALARM	Component Point not in alarm.
Color is Reverse	ALARM	Component in alarm "Emergency" = Magenta Component in alarm "High" = Red Component in alarm "Low" = Yellow

9. Schematic Operation

9.1. Stream Status Displays

PV Low Limit

The Station must be in Engineering level. To enter a new PV low limit (PVLOTP) for a component, select the PVLOTP value for the component. The display will prompt the user to enter a new low limit.

PV High Limit

The station must be in Engineering level. To enter a new PV high limit (PVHITP) for a component, select the PVHITP value for the component. The display will prompt the user to enter a new high limit.

Command Target Action

The action associated with the COMMAND 2ND-STD target is dependent upon the Command Code Requested.

Table 9-6 Command Target Action

Command Code Requested	CDS Parameter COMMVAL	Target Action
	-1	Target does not appear on display
	0	User is prompted to enter a Command Code to be sent
	Not 0	Command Code Requested will be sent

Stream Detail Display

To invoke the Detail Display for the Stream, select the Stream point.

Return to First Level

To return to the first level display, select the DISPLAY BACK target.

Additional Display Targets

When a stream component is selected, two additional targets appear on the Stream Component Status display.

Figure 9-3 Stream Component Status Display

DELETE POINT OR DETAIL

07 Nov 94 15:14:22

CLM/AA H

CLM/AA HCI A COMPONENT STATUS

DESC: ANALYZER 1 - STREAM

CYCLE : 00:15

POINT: A01STR01

DETECTOR : 5

DETECTOR STATE :

STREAM :

STATUS :

ALM CODE :

LAST RESUT : 10:30

NEXT UPDATE : 10:30

COMPONENT	DESCRIPTION	EU	PVLOTP	PVHITP
12.500000	COMPONENT 1	PERCENT	10.000000	15.000000
0.0200000	COMPONENT 2	PERCENT	0.0010000	0.0500000

STR1_COM

DELETE
POINT

DETAIL
DISPLAY

SEND TIME
AND DATE

COMMAND
2ND-STD

DISPLAY
BACK

Component Detail Display

To invoke the Detail Display for a component point, first select the component value, then select the DETAIL DISPLAY target. The Detail Display for the component point will then be displayed.

If a component point does not exist for the component slot, the display will prompt the user to enter a new component point. This must be the name of an existing stream component point (AM regulatory).

9. Schematic Operation

9.1. Stream Status Displays

Deleting a Component

The Station must be in Engineering level. To delete a component point from the display, first select the component value, then select the DELETE POINT target. The component point will then be cleared from the display.

10. Restarting the APP Node

The following steps are done to reset an APP node. The following subsections define the steps to reset an APP node.

- a) Power down the APP node. Only do this when the COM port on the APP node is not responding. If the COM port is responding then go to the next step of resetting the LCNP card.
- b) Reset the LCNP card.
- c) Load the AM and CLM databases.
- d) Start the HCIA interface.

10.1 From a Power Down State

Power up and make sure the 'TDC Emulator Service' has started

The APP node power down only needs to be done when the COM ports are not responding. When this happens, power down for a full two minutes.

Power up the APP node. Verify that the TDC Emulator Service has started. If not start the service.

10.2 Reset the LCNP

Reset the LCNP card

Verify that the 'TDC Emulator Service' has started.

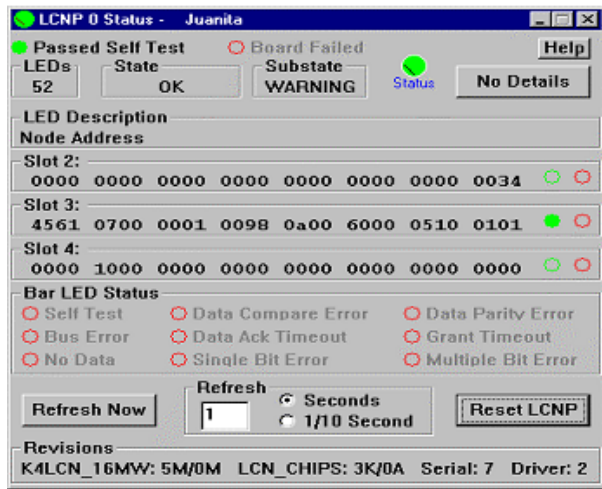
Maximize the LCNP window. Select the Detail button if the LCNP Status window does not show the details. Refer to Figure 10.1, LCN Status Window.

Select the "Reset LCNP" button to reset the LCNP card. The LCNP card will be reset.

The AM and CLM databases need to be reloaded once the LCNP card has been reset.

10. Restarting the APP Node
10.3. Reload the AM and CLM databases

Figure 10-1 LCNP Status Window



10.3 Reload the AM and CLM databases

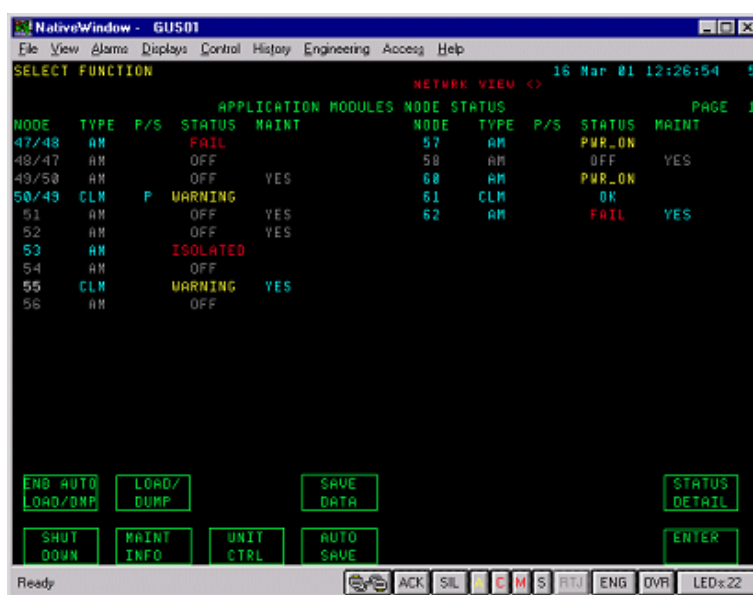
Verify that the 'TDC Emulator Service' has started.

From the LCN System Status Native window select the node then select 'NODE STAT'. The 'Application Module Node Status' window is displayed. Refer to Figure 10.2. Select the load type. This loads the AM and CLM databases. The node status goes to TEST, QUALIF, NET LOAD, READY and finally to OK or WARNING once the AM and CLM databases are loaded.

10. Restarting the APP Node

10.4. Restart the HCIA

Figure 10-2 Services Window



10.4 Restart the HCIA

Reset the HCIA communications

The HCIA card that the CLM communicates with may have gone into an error condition that causes the HCIA card to not send data to the CLM. Reset the HCIA card if this has happened.

10. Restarting the APP Node

10.4. Restart the HCIA

11. Terminology and Definitions

11.1 Terms Used in This Document

Terms Used

CLM

A CLM is defined as a Communication Link Module. A CLM may be hosted on an AM (AM/CLM) or an APP (APP/CLM).

AM/CLM

An AM/CLM is defined as a CLM hosted on an Application Module (AM) with a least one CLMI board and the CLM software package installed. The AM must be R600 (or later) revision with at least 4 megawords of memory.

APP/CLM

A APP/CLM is defined as a CLM hosted on an APP node. A APP/CLM utilizes the TDC Emulator Service to emulate the functionality of the CLMI board.

Image Point

An operating point in the CLM images one analyzer stream from an Applied Automation analyzer.

Network Device

Network Device refers to the non-TPS device or network that interfaces to the TPS by way of the CLM.

Network Memory

Network Memory is the portion of CLM memory defined for shared access with the CLMI board to perform the network communication task.

11. Terminology and Definitions

11.2. HCI-A Interface Package

11.2 HCI-A Interface Package

Deliverables for the CLM/Applied Automation HCI-A Interface Package

The CLM/Applied Automation HCI-A Interface Package for an AM based CLM includes the following items:

Documents:

CLM/Applied Automation HCI-A Interface Manual, (this document).

Object Modules:

- Communications Protocol Software - One Object Module (AAHCIA.S)
- Control Language (CL) Object Programs to support device or network specific requirements.

Control Language Files:

- Custom Data Segment (CDS) Source Packages for custom AM/CLM image points.
- Custom Display Package for device or network specific functions.
- Unlink(UNLK)/Link(LK) CL Blocks to an image point

Template Definition:

- Template files for the supported NDT layer
- PITE template file for each image point type
- Prototype BIFs for each template type

Installation Files:

- CLMI Configuration File for Applied Automation HCI-A Interface

Separate Interface Installation command files to:

- Copy files to appropriate HM directories
- Compile all CDS packages included by Honeywell.

Deliverables for the APP/CLM Applied Automation HCI-A Interface Package

The APP/CLM Applied Automation HCI-A Interface Package for an APP based CLM includes all items of the CLM/Applied Automation HCI-A Interface Package in addition to the following items:

Windows Modules:

- AAHCIA.DLL
- HCIMSG.DLL

Windows Installation Files:

- CLM AAHCIA Install.exe for Windows 2003
- CLM_AAHCIA_Install.msi for Windows 2008

Other Reference Documents



REFERENCE - INTERNAL

CLM Platform Implementation Manual, Document LK12-400, dated 03/11.



REFERENCE - EXTERNAL

Applied Automation Optichrom Computer Communication Protocol Manual, Document 20006515-008 January, 1991.

11. Terminology and Definitions

11.2. HCI-A Interface Package

12. Appendix A – Analyzer Data

Example of a link level data

CLMI	HCIA	Comments
ENQ		
	ACK	Line check between CLMI and HCIA card
	STX	
	1.0,	Analyzer number
	130.0,	Transmission of results message
	946.0,	Time
	14.0,	Day
	12.0,	Month
	94.0,	Year
	1.0,	Stream number
	1.0,	Analyzer number
	1.0,	Analyzer status
	0.0,	Alarm code
	40.475	Result #1
	41.475	Result #2
	42.475	Result #3
	43.475	Result #4
	44.475	Result #5
	45.475	Result #6
	46.475	Result #7
	47.475	Result #8

12. Appendix A – Analyzer Data

11.2. HCI-A Interface Package

	48.475	Result #9
	49.475	Result #10
	.	.
	.	.
	49.475	Result #39
	49.475	Result #40
	ETX	
	A2	Checksum
STX		
1.0,		Analyzer number
128.0		OK message - Message received correctly
ETX		
E6		Checksum

13. Appendix B – Alarm Messages

Table of Alarm Messages

Summary of alarm messages generated by the CL that is attached to image point.

STREAM POINT DETAIL POINT STATE ALARM DESC	COMPONENT POINT PV VALUE	CLM/HCIA COMPONEN T STATUS STATUS VALUE	COMMENTS
OFFLINE OFFLINE	PV set to bad value.	OFFLINE	CLMI device status is set to OFF.
OFFLINE OFFLINE	PV set to bad value.	OFFLINE	Point has just been set to active. This state should go away in a short period of time.
WARNING WARNING	Current PV value.	WARNING	An alarm value between 0 and 127 has been received.
WARNING TIME OUT	PV set to bad value.	WARNING	The CL watchdog timer has detected that no new data has been received from the HCIA in the TIMEOUT value specified by the operator.
WARNING # COMP	PV set to bad value.	WARNING	Mismatch of number of components specified by the user and number of components specified in the stream data from the HCIA. Refer to NUMCOMP and NUMSTMCM parameters in the HCIASTRM CDS package.

13. Appendix B – Alarm Messages

11.2. HCI-A Interface Package

INVALID FAULT	PV set to bad value.	INVALID	An alarm value between 128 and 255 has been received
LIMIT COMP ALM	PV set to bad value.	LIMIT	An associated component point has gone into alarm. Refer to the component points.

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